



University of
Nottingham

UK | CHINA | MALAYSIA

A large, high-resolution image of the Earth as seen from space, showing the Western Hemisphere with North and South America. The image is framed by a thin white border.

COMP-2024 / COMP-2039
**Artificial Intelligence
Methods**

Lecture 10
Knowledge and Reasoning



Learning Outcomes

IDENTIFY

1. Types of Knowledge
2. Representation Methods
3. Case-Based Reasoning
4. Decision Making, Decision Support
5. LLMs in Knowledge Representation and Reasoning



ACK: Prof. Ender Özcan, UNUK, Prof. Tomas Maul & Dr Chen ZhiYuan, UNM



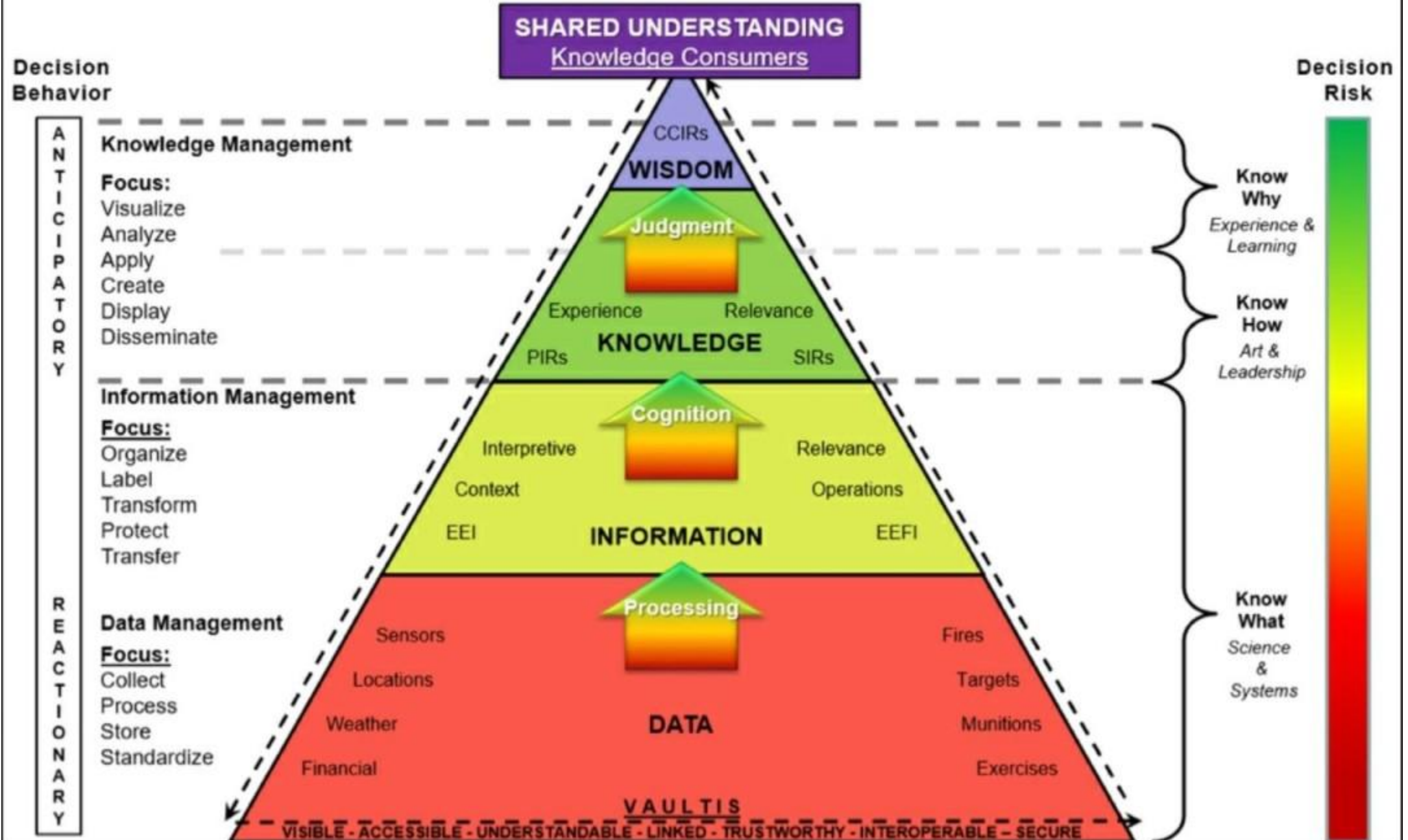
AI and Knowledge Representation & Reasoning

- A description of **Artificial Intelligence** is:
 - The study and development of systems that demonstrate **intelligent behavior**.
- Based on the above, a description of **Knowledge Representation & Reasoning** is:
 - The study of ways to **represent** and **reason** with **information** in order to achieve *intelligent behavior*.



KR&R is the part of AI that is concerned with **thinking (cognition)** and how thinking contributes to intelligent behavior.

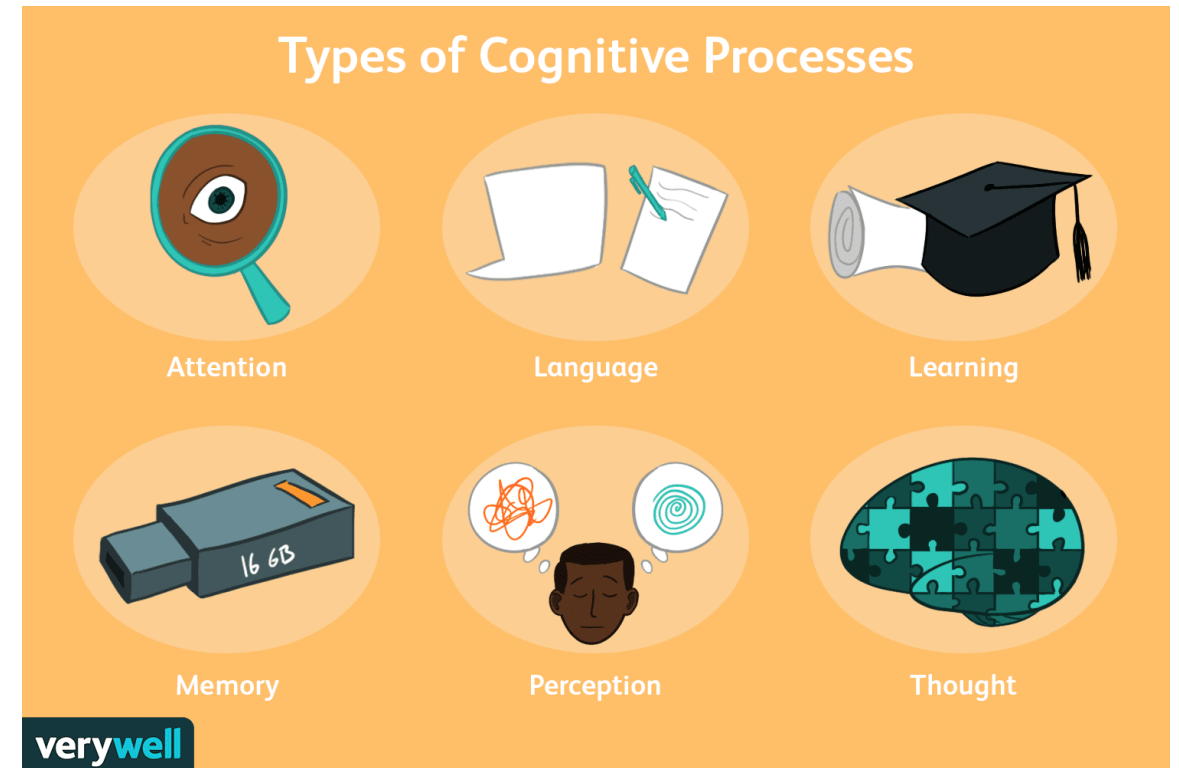
The KM Cognitive Pyramid





Cognition

- **Cognition** refers to the **mental processes** involved in acquiring, processing, storing, and using information. These processes include perception, attention, memory, reasoning, problem-solving, and decision-making.
- Cognition is what allows humans to understand and interact with the world, learn from experiences, and adapt to new situations.



ACK: Prof. Ender Özcan, UNUK, Dr Tomas Maul & Dr Chen ZhiYuan, UNM



What is Knowledge Representation & Reasoning (KR&R)?

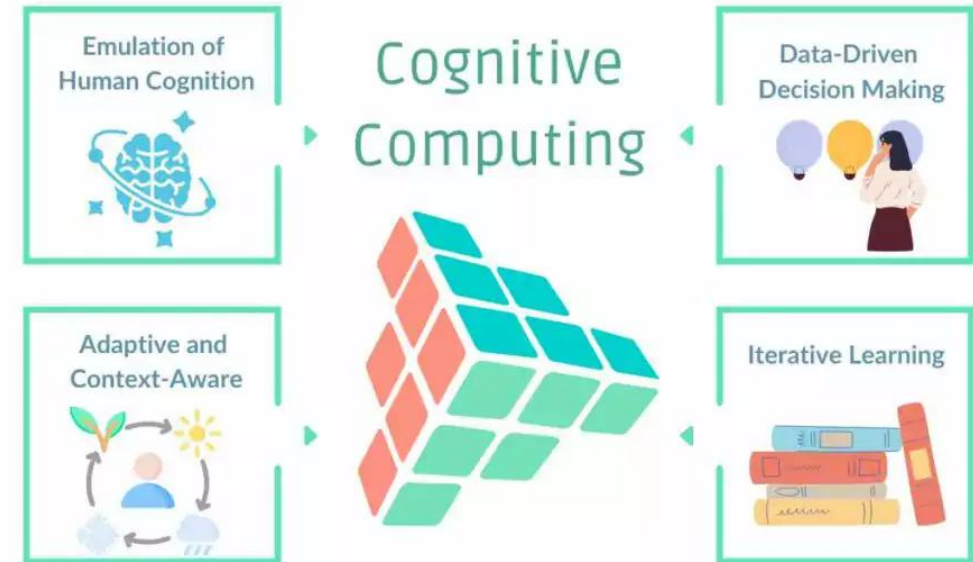
- **KR** suggests an approach to understanding intelligent behavior that is radically different.
 - Instead of studying humans very carefully (biology, nervous systems, psychology, sociology, etc.), it argues that what we need to study is **what humans know (knowledge structure)**.
 - It is taken as a given that what allows humans to behave intelligently is that they ***know** about a lot of things* and are able to *apply this **knowledge*** as appropriate to adapt to their environment and achieve their goals.
- **KR&R focuses on the *knowledge***, not on the knower. We ask what any agent - human, animal, electronic, mechanical - would need to **know** to behave intelligently, and what sorts of computational mechanisms might allow its knowledge to be manipulated.

ACK: Prof. Ender Özcan, UNUK, Prof. Tomas Maul & Dr Chen ZhiYuan, UNM



Cognitive Computing

- **Cognitive computing** is a branch of computing that aims to mimic human cognitive abilities using artificial intelligence (AI) and machine learning (ML). It involves systems that can learn, reason, understand natural language, and interact with humans in an *intelligent* way.
- Unlike traditional computing, which follows strict rule-based logic (programming), cognitive computing systems analyze vast amounts of data, recognize patterns, and make predictions or recommendations.



ACK: Prof. Ender Özcan, UNUK, Dr Tomas Maul & Dr Chen ZhiYuan, UNM



Knowledge

What is knowledge? This is a question that has been discussed by philosophers since the ancient times, and it is still not totally clarified.

- Observe that when we say something like “We know that ...,” we fill in the blank with a simple
 - “I know that Tomas will not come to today's class”
 - “We know that Tomas is the module convenor”
- Among other things, **knowledge is a relation between a knower and a proposition**
 - **knower** : I
 - **proposition**: the idea expressed by a simple declarative sentence, like “Tomas will not come to today's class.”
- What can we say about propositions? For KR&R, what matters about **propositions** is that they are abstract entities that can be *true* or *false*, right or wrong.
 - When we say, “We know that p,” we can just as well say, “We know that it is true that p.”

ACK: Prof. Ender Özcan, UNM, Prof. Tomas Maul & Dr Chen ZhiYuan, UNM



Types of Knowledge

- **Declarative Knowledge:** Facts and rules about the world. (What)
- **Procedural Knowledge:** Knowledge of how to perform tasks or procedures. (How)



Types of Knowledge

- **Declarative Knowledge**

- Description of *notions*, *facts*, and *rules* of the world.
 - E.g.
 - For each lecture there is a specific time and place.
 - Only one lecture can take place at each time and place.
- Descriptive knowledge, non-procedural, independent of targets and problem solving.



Types of Knowledge

- **Procedural Knowledge**

- Description of *procedures* required to achieve targets.
- Knowledge of the *order* in which actions must be performed.
- Heuristic knowledge.

- E.g.

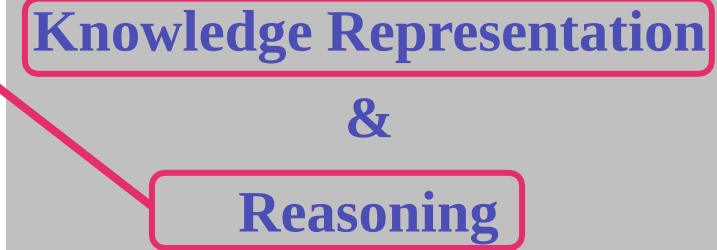
- To construct the exams timetable, assign first the classes of the first year.
- To reach Beijing faster, take the airplane.

- It depends on the *targets* and *problems*.



Types of Knowledge in KR&R

- **Declarative knowledge** is *right* or *wrong*. (What)
 - Lectures are on Thursday.
- **Procedural knowledge** can be *executed*. (How)
 - the procedure of constructing the exams timetable.





Declarative Knowledge vs Procedural Knowledge

Aspect	Declarative Knowledge	Procedural Knowledge
Definition	Facts, concepts, or rules about the world.	Knowledge of how to perform tasks or procedures.
Nature	Descriptive: Focuses on "what is" or "what is true."	Action-oriented: Focuses on "how to" do something.
Examples	- The capital of France is Paris. - Water boils at 100°C at sea level.	- How to ride a bicycle. - How to solve a math problem using a specific method.
Format	Stored in statements or propositions.	Embedded in processes, algorithms, or sequences.
Retrieval	Easy to verbalize and explain explicitly.	Often implicit and harder to explain without doing.
Learning Process	Acquired by studying or memorization.	Acquired by practice and repetition.
Application	Useful for recalling facts or making decisions.	Useful for performing actions or solving problems.
Storage in the Brain	Often associated with the hippocampus .	Often associated with the cerebellum and motor areas .
Dependence	Can exist independently of procedural knowledge.	Often builds on declarative knowledge for context.

ACK: Prof. Ender Özcan, UNUK, Dr Tom



Representation

- AI agents deal with knowledge (not data).
 - **Facts** (believe & observe knowledge).
 - **Procedures** (how to knowledge).
 - **Meaning** (semantics) (relate & define knowledge).
- Right representation is crucial.
 - Early realisation in AI.
 - Wrong choice can lead to project failure.
 - Active research area.
- **Knowledge representation**, then, is the field of study concerned with using *formal symbols* to represent a collection of propositions believed by some agent.

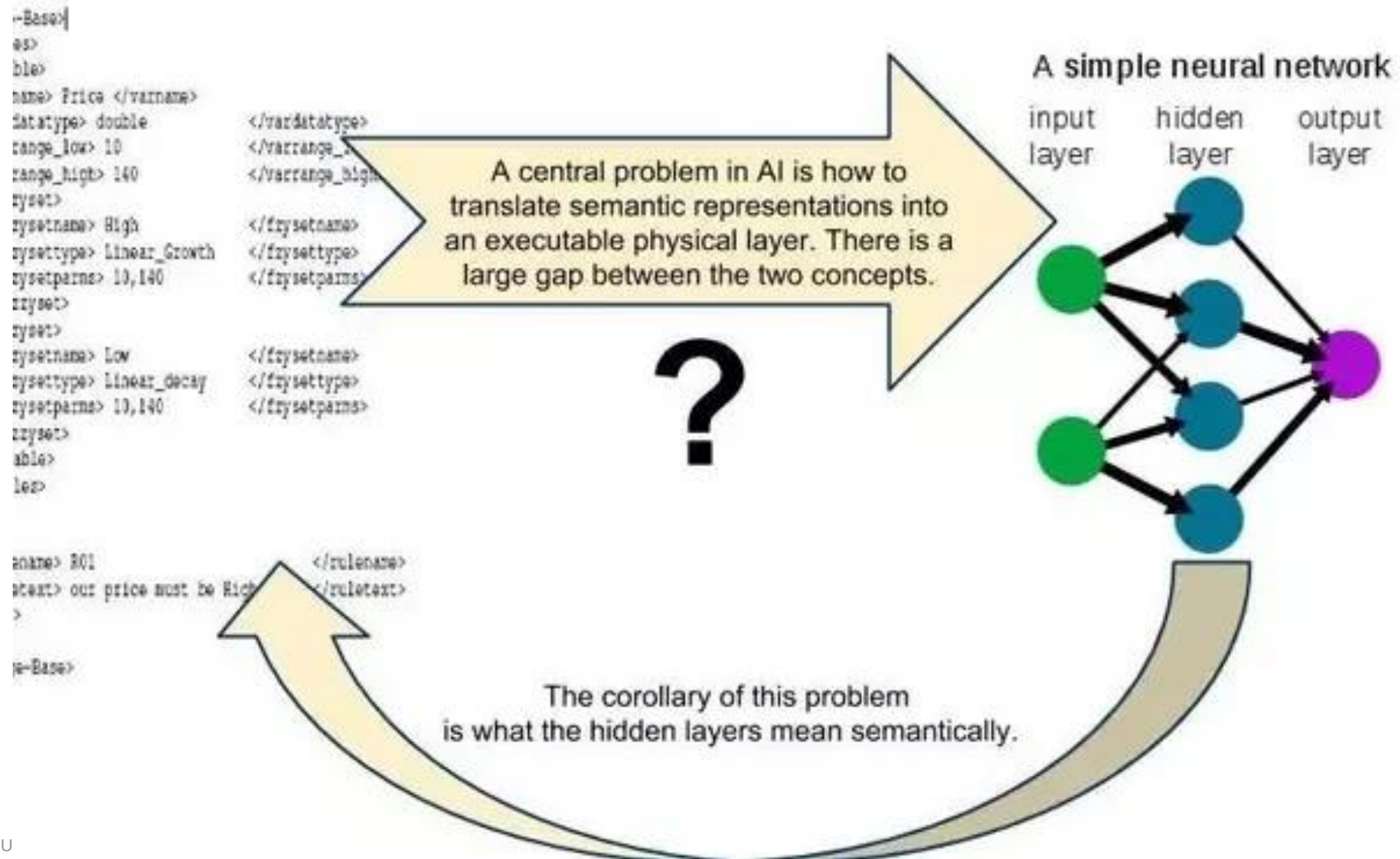


Knowledge Representation

- **Symbolic representation** relies on explicitly defined *structures and rules*, making it suited for problems requiring logical reasoning.
 - Year 3: Symbolic Artificial Intelligence (COMP3070)
- **Non-symbolic representation** is *data-driven* and uses statistical or neural methods, excelling in areas like perception and recognition. (machine learning)



Knowledge Representation



ACK: Prof. Ender Özcan, U



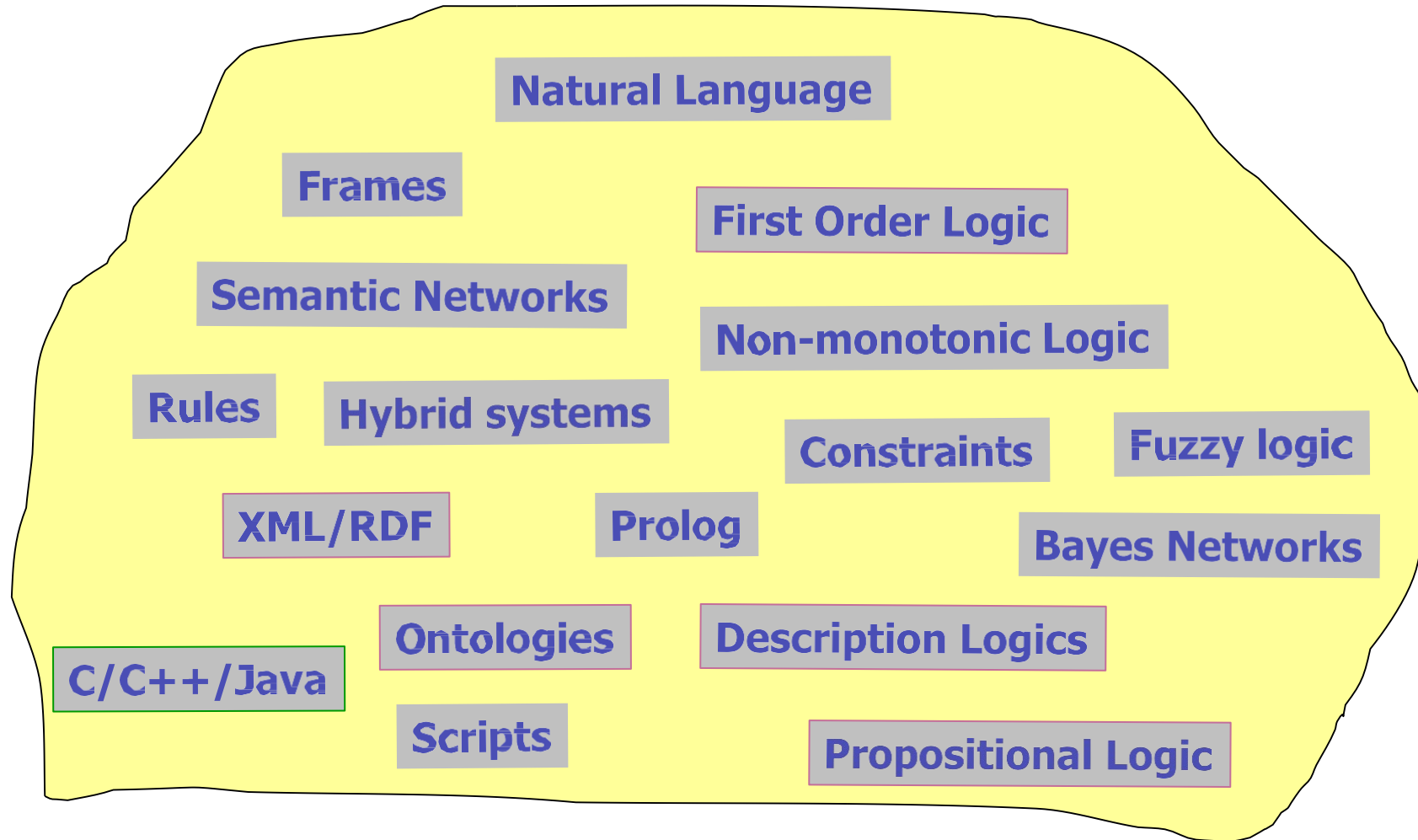
Symbolic Methods

- **Symbolic methods** represent knowledge using formal, *rule-based*, and *logical* representations. Knowledge is encoded using *symbols* (such as words, numbers, or logical expressions) that represent abstract concepts, entities, and their relationships.
- **Top Down** approach with pre-defined structure
- **Symbolic reasoning** involves the manipulation of these symbolic representations using *formal rules* and algorithms to draw conclusions or make inferences.
- Examples of symbolic methods include *logic-based representations*, *semantic networks*, and *rule-based systems*.

ACK: Prof. Ender Özcan, UNUK, Dr Tomas Maul & Dr Chen ZhiYuan, UNM



Symbolic Methods of Knowledge Representation



ACK: Prof. Ender Özcan, UNUK, Prof. Tomas Maul & Dr Chen ZhiYuan, UNM



Representation Methods (Symbolic)

1. Logical Representations
 - Propositional Logic
 - Predicate logic (first-order logic)
 - Higher Order Logic
 - Multi-valued logics
 - Modal logics
2. Production Rules
3. Semantic Networks
 - Conceptual graphs
 - Frames



Non-Symbolic Methods (Neuro / Sub-symbolic)

- **Non-Symbolic / Subsymbolic / Neuro AI** refers to approaches in artificial intelligence that do not rely on explicit symbolic representations or rules but instead use distributed, numerical, and statistical methods to process information. These methods are inspired by how biological systems, such as the human brain, learn and operate.
- **Bottom-up** approach without pre-defined structure
- Knowledge is encoded in the form of *numerical weights, activations, and connections* between processing units (neural network), rather than explicit symbolic representations.
- **Non-symbolic reasoning** involves the processing of these distributed representations, often through machine learning techniques such as neural networks and deep learning.
- **Examples of non-symbolic methods** include connectionist models, distributed representations, and probabilistic graphical models.

ACK: Prof. Jorge Ozcariz, University of Toronto, & Dr. Chen Shiyuan, UNIM



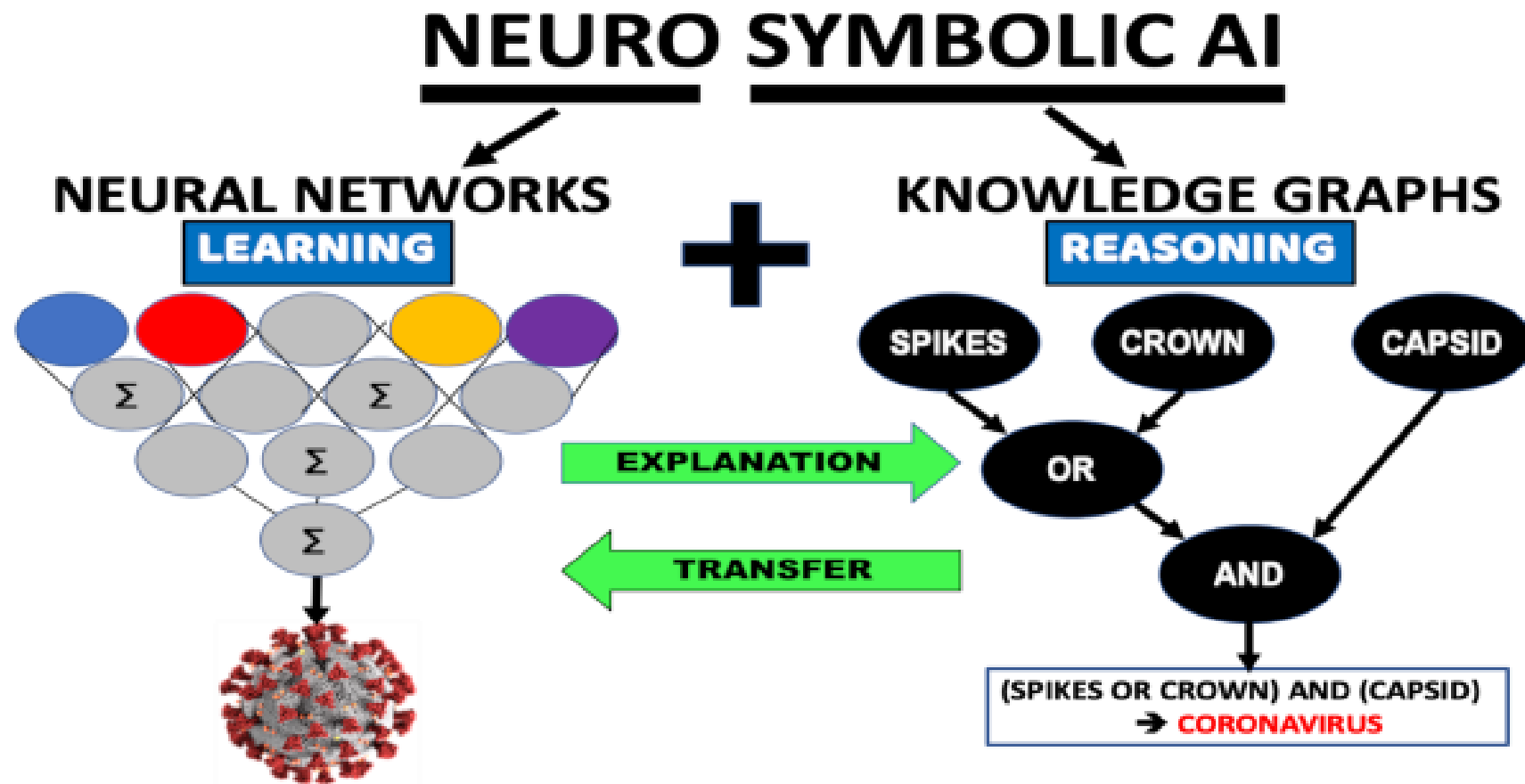
Non-Symbolic Methods (Neuro / Sub-symbolic)

- **Bottom-up Learning:** Knowledge is acquired through learning from data rather than being explicitly programmed.
- **Distributed Representations:** Information is encoded across many units (e.g., neurons in a neural network).
- **Adaptive and Flexible:** Learns patterns, relationships, and features from raw data.
- **Handles Uncertainty and Noise Well:** Effective in dealing with complex, ambiguous, or unstructured data.

ACK: Prof. Ender Özcan, UNUK, Dr Tomas Maul & Dr Chen ZhiYuan, UNM



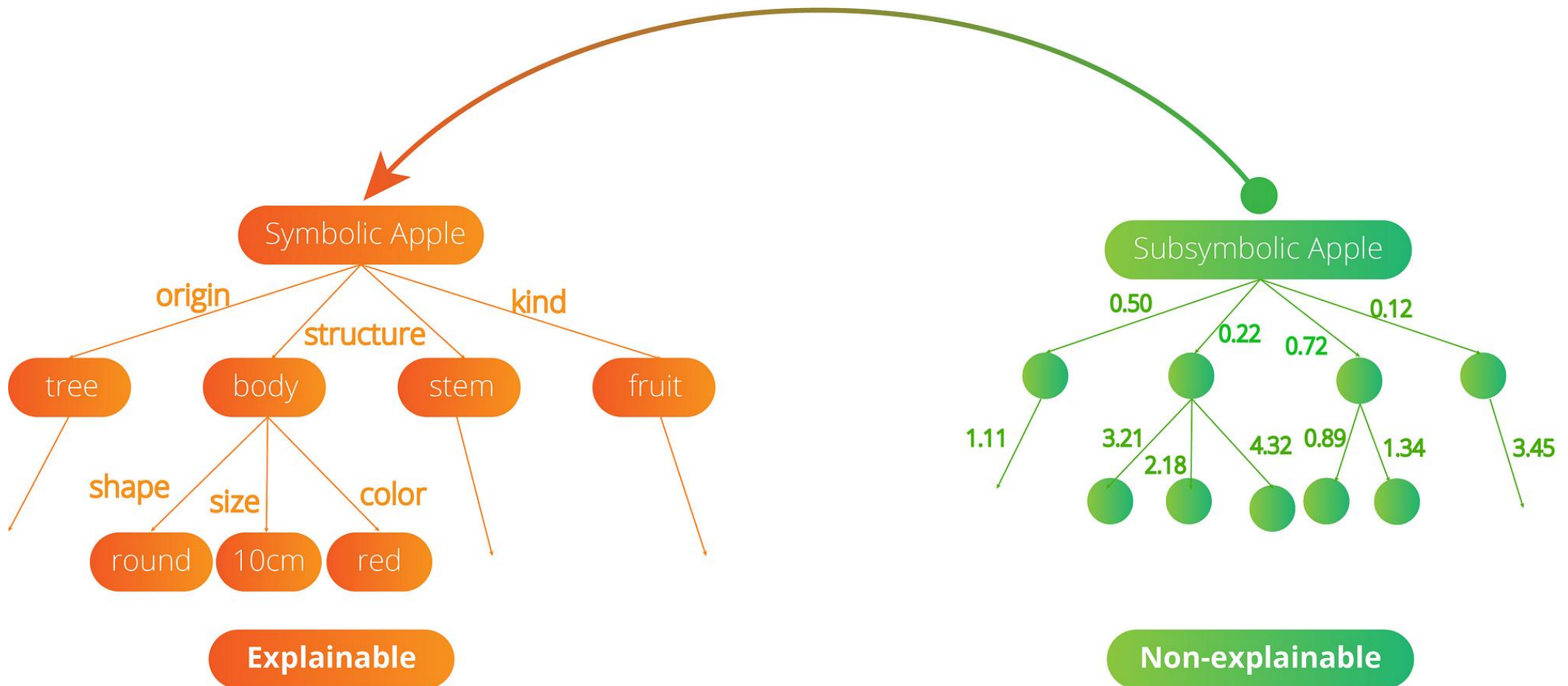
Neuro vs Symbolic AI



ACK: Prof. Ender Özcan, UNUK, Dr Tomas Maul & Dr Chen ZhiYuan, UNM



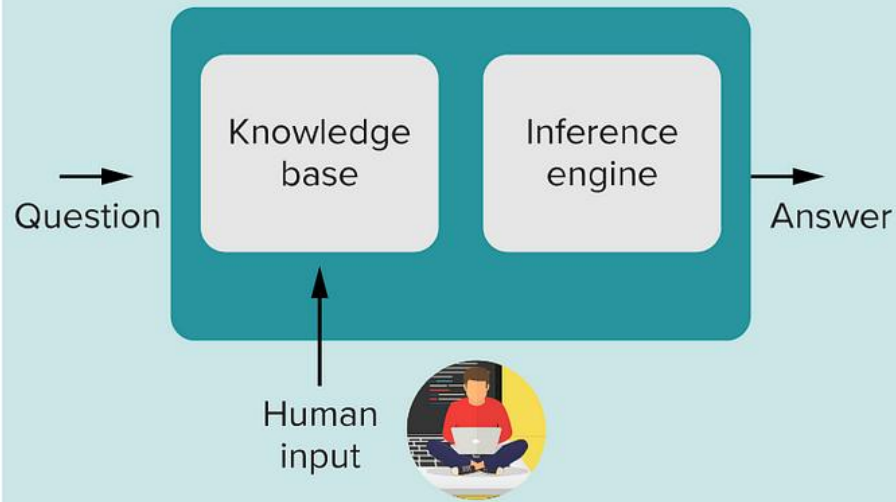
Neuro vs Symbolic AI



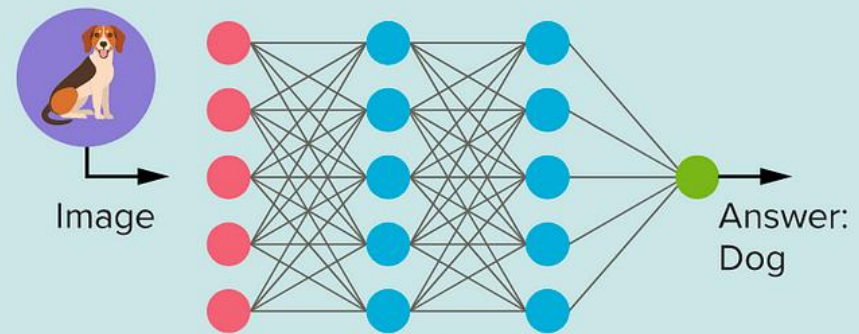
ACK: Prof. Ender Özcan, UNUK, Dr Tomas Maul & Dr Chen ZhiYuan, UNM



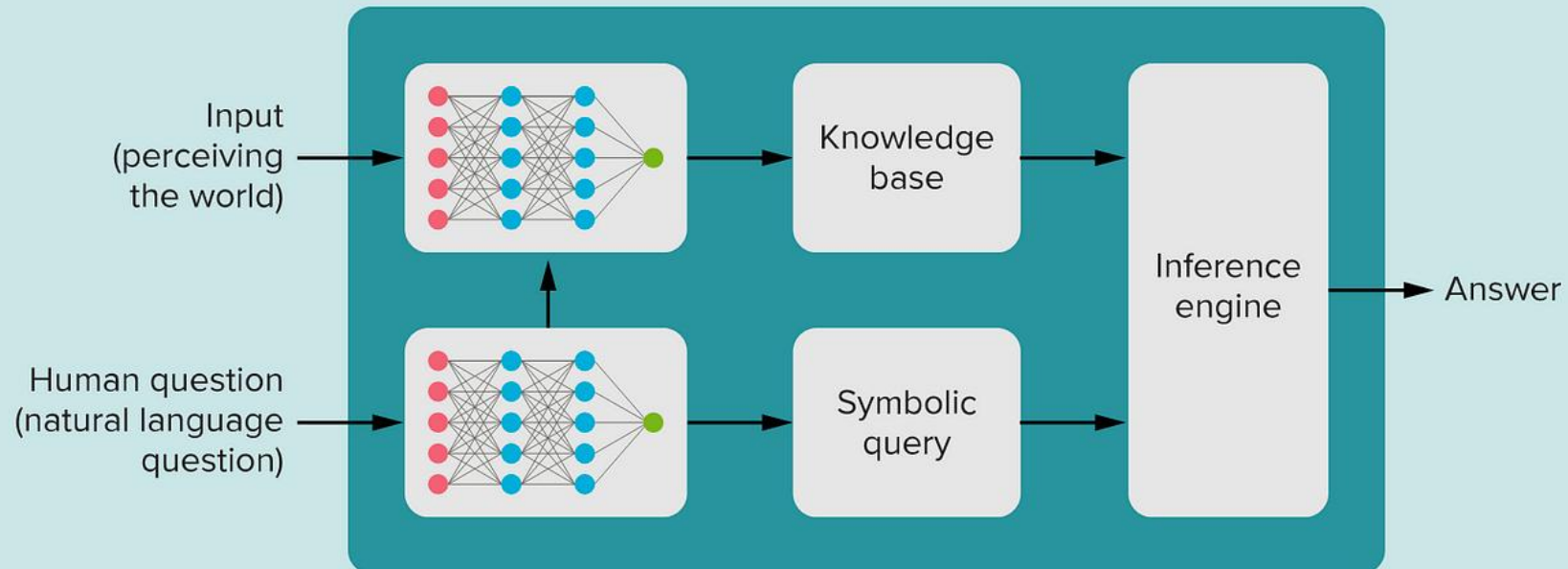
SYMBOLIC AI



DEEP NETS



HYBRID AI





Knowledge Representation

Aspect	Symbolic Knowledge Representation	Non-Symbolic Knowledge Representation
Definition	Represents knowledge using explicit symbols, such as words, rules, or logic.	Represents knowledge implicitly, often using patterns, weights, or connections.
Nature	Discrete and rule-based.	Continuous and pattern-based.
Format	Logical statements, ontologies, semantic networks, or production rules.	Neural networks, statistical models, or distributed representations.
Examples	- "If-then" rules (e.g., If it rains, then carry an umbrella).	- Recognizing a face in an image through a neural network.
Interpretability	Highly interpretable: easy to understand and explain.	Often opaque or difficult to interpret (e.g., "black box" models).
Processing Approach	Top-down, based on explicit reasoning or logical inference.	Bottom-up, based on data-driven learning and pattern recognition.
Flexibility	Rigid, as it requires precise definitions and structure.	Adaptive, as it can generalize and handle ambiguity.
Learning Process	Typically requires human encoding of knowledge.	Learned automatically from data through training.
Storage	Stored as symbols, rules, or structured databases.	Stored in distributed forms, such as neural weights or probabilities.
Application Domains	- Expert systems (e.g., medical diagnosis systems).	- Machine learning (e.g., image recognition, speech processing).
Advantages	- Easy to debug and modify.	- Robust to noisy or incomplete data.
Disadvantages	- Struggles with ambiguity or incomplete information.	- Hard to interpret and understand its internal reasoning.

ACK: Prof. Ender Özcan, UNUK, Dr Tomas Maul & Dr Che



How can knowledge be represented?

Symbolic

- Declarative Languages (Logic)
- Imperative Languages (C, C++, Java, etc.)
- Hybrid Languages (Prolog)
- Rules
- Frames
- Semantic Networks
- ...

Non-Symbolic

- Connectionist Representations
 - Neural networks
 - Deep learning models
- Probabilistic Graphical Models:
 - Bayesian networks,
- Genetic Algorithms (GAs)
- Reinforcement Learning (RL)

ACK: Prof. Ender Özcan, UNUK, Dr Tomas Maul & Dr Chen ZhiYuan, UNM



University of
Nottingham

UK | CHINA | MALAYSIA

Logical Representations



What is a Logic?

- A language with **concrete rules**.
 - No ambiguity (discrete, deterministic) in representation (may be other errors!).
 - Allows *unambiguous* communication and processing.
 - Very unlike natural languages e.g., English.
- Many ways to translate between languages
 - A statement can be represented in different logics.
 - And perhaps differently in same logic.
- *Expressiveness* of a logic.
 - Which description is more accurate/expressive?
- Not to be confused with *logical reasoning*.
 - Logics (nouns) are languages, reasoning (verb) is a process (may **use** logic).

ACK: Prof. Ender Özcan, UNUK, Prof. Tomas Maul & Dr Chen ZhiYuan, UNM

Logical Representation: Combining Quantifiers

Every student has a dog.

"For every x , if x is a student, then there exists a y such that y is a dog and x owns y ."

$$\forall x (Student(x) \rightarrow \exists y (Dog(y) \wedge Owns(x, y)))$$

Spanish

Cada estudiante tiene un perro.

Chinese

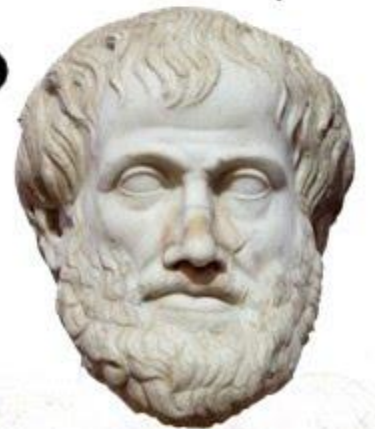
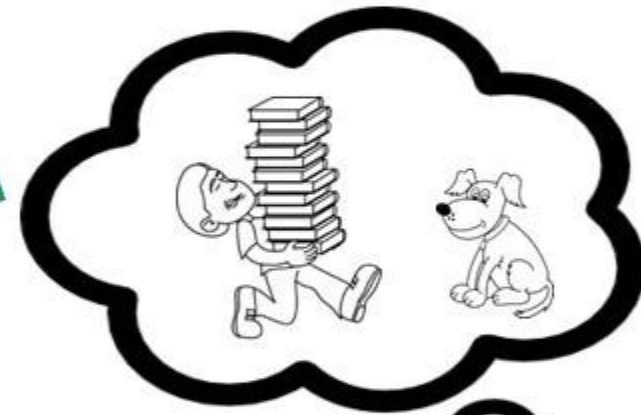
学生们都有狗

Bulgarian

Всеки ученик има куче.

Greek

κύων πᾶσι τοῖς μαθηταῖς ὑπάρχει





Syntax and Semantics

- Before any system aspiring to intelligence can even begin to reason, learn, plan, or explain its behavior, it must be able to *formulate the ideas* involved.
 - You will not be able to learn something about the world around you, for example, if it is beyond you to even **express** what that thing is.
- So, we need to start with a **language**, in terms of which knowledge can be formulated.
- **What does it mean to “have” a language?** Once we have a set of words or a set of symbols of some sort, what more is needed? As far as we are concerned, there are two main things:
 - A KR language is defined by its **syntax** and its **semantics**.

ACK: Prof. Ender Özcan, UNUK, Prof. Tomas Maul & Dr Chen ZhiYuan, UNM



Syntax

- We need to specify which groups of symbols, arranged in what way, are to be considered properly formed.
- In English, for example, the string of words “the cat my mother loves” is a well- formed phrase, but “the my loves mother cat” is not.
- The **syntax** consists of a set of **symbols** used by the language and a set of **rules (grammar)** according to which the symbols can be combined to form proper sentences.



Semantics

- We need to specify what the well-formed expressions are supposed to **mean**.
 - Some *well-formed expressions* like “the recently accepted decimal holiday” might not mean anything. We need to be clear about what idea about the world is being expressed.
- The **semantics** determine a **mapping** between symbols, combinations of symbols, propositions of the language and *concepts of the world* to which they refer.
- A proposition in a KR language does not mean anything on its own.
 - The **semantics** (i.e., the meaning) of the proposition must be defined by the language author through an **interpretation**.

ACK: Prof. Ender Özcan, UNUK, Prof. Tomas Maul & Dr Chen ZhiYuan, UNM



Syntax and Semantics

- Syntax
 - **Rules** for constructing legal sentences in the logic.
 - Which **symbols** we can use (English: letters, punctuation).
 - How we are allowed to **combine** symbols.
- Semantics
 - How we **interpret** (read) sentences in the logic.
 - Assigns a **meaning** to each sentence.
- Example: “**All lecturers are female**”
 - A valid sentence (syntax).
 - And we can understand the meaning (semantics).
 - This sentence happens to be **false** (there is a counterexample).

ACK: Prof. Ender Özcan, UNUK, Prof. Tomas Maul & Dr Chen ZhiYuan, UNM



Predicate Logic – Syntax and Semantic

- **Syntax:** $\forall x(\text{Lecturer}(x) \rightarrow \text{Female}(x))$ → with object x and predicate $\text{Lecturer}(x)$, $\text{Female}(x)$
- **Semantic:** "For all x , if x is a lecturer, then x is female."



Propositional Logic – Syntax and Semantic

- **Syntax:** $L \rightarrow F$
 - L represent "A person is a lecturer."
 - F represent "A person is female."
- **Semantic:** "If someone is a lecturer, then they are female."
- In propositional logic, we deal with whole statements (propositions) rather than individual objects and predicates.



Syntax vs Semantic

Syntax

- **Syntax** is the formal, grammatical structure of the knowledge representation language.
- It defines the rules and formatting for how knowledge is expressed and encoded.
- Common knowledge representation syntaxes include:
 - Predicate logic (e.g., first-order logic)
 - Frame-based systems (e.g., RDF, OWL)
 - Rule-based systems (e.g., production rules)
 - Semantic networks
 - Ontologies

Semantics

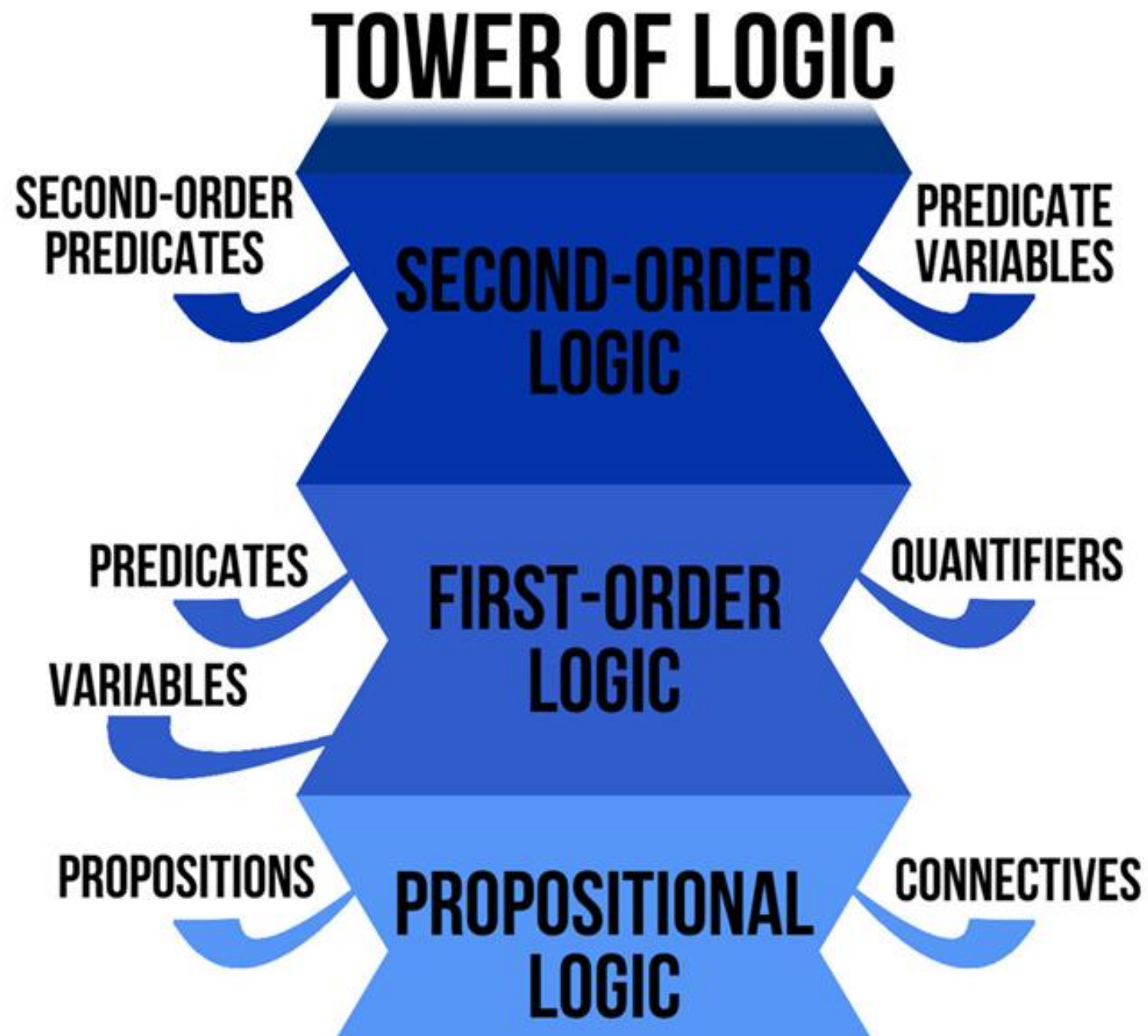
- **Semantics** refers to the meaning and interpretation of the knowledge representation.
- It defines the interpretation and reasoning mechanisms associated with the knowledge.
- Semantics are crucial for enabling machines to understand and reason about the knowledge.
- Examples of semantic aspects include:
 - Concept definitions and relationships
 - Inference rules and reasoning mechanisms
 - Logical entailment and consistency
 - Ambiguity and uncertainty handling



University of
Nottingham

UK | CHINA | MALAYSIA

Propositional and Predicate Logic (First Order Logic)



ACK: Prof. Ender Özcan, UNUK, Dr Tomas



Propositional Logic

- **Propositional logic**, also known as sentential logic, is a formal system of logic that deals with propositions or statements that can be either true or false.
- The basic building blocks are atomic propositions (e.g., "The sky is blue") and logical connectives (e.g., AND, OR, NOT, IF-THEN).
- **Propositional logic** is widely used in various applications, such as:
 - Computer programming and software engineering (e.g., Boolean algebra, digital circuit design)
 - Artificial intelligence and knowledge representation (e.g., expert systems, automated reasoning)
 - Mathematical logic and formal verification (e.g., theorem proving, model checking)
 - Natural language processing (e.g., semantic analysis, logical inference)



Propositional Logic

- Syntax

- **Atomic Propositions**, e.g., “it is wet”.

- **Connectives**: and, or, not, implies, if and only if

$\wedge \quad \vee \quad \neg \quad \longrightarrow \quad \longleftrightarrow$

- **Brackets**, T (true) and F (false)

- Semantics (Classical a.k.a. Boolean)

- Define how connectives affect truth.

- “P and Q” is true if and only if P is true and Q is true.

- Use **truth tables** to work out the truth of statements.

ACK: Prof. Ender Özcan, UNUK, Prof. Tomas Maul & Dr Chen ZhiYuan, UNM



Propositional Logic Connectives



Connective Symbols	Words	Terms	Example
$\wedge$	AND	Conjunction	$A \wedge B$
$\vee$	OR	Disjunction	$A \vee B$
$\rightarrow$	Implies	Implication	$A \rightarrow B$
$\Leftrightarrow$	If and only If	Biconditional	$A \Leftrightarrow B$
$\neg$	Not	Negation	$\neg A$



First Order Logic (FOL)

- **First-order logic**, also known as **predicate logic**, is a more expressive and powerful formal system of logic compared to propositional logic.
- **Predicate**: a function returning true/false
- It allows for the representation and reasoning about individuals, their properties, and the relationships between them.



Key components and characteristics of first-order logic

- **Constants** are objects: john, apples
- **Predicates** are properties and relations:
 - likes(john, apples)
- **Functions** transform objects:
 - likes(john, fruit_of(apple_tree))
- **Variables** represent any object: likes(X, apples)
- **Quantifiers** qualify values of variables
 - True for all objects (Universal): $\forall X. \text{likes}(X, \text{apples})$
 - Exists at least one object (Existential): $\exists X. \text{likes}(X, \text{apples})$

ACK: Prof. Ender Özcan, UNUK, Dr Tomas Maul & Dr Chen ZhiYuan, UNM



Propositional Logic vs First Order Logic

Propositional Logic

- Propositional logic deals with propositions or statements that can be either true or false.
- The basic building blocks are atomic propositions (e.g., "The sky is blue") and logical connectives (e.g., AND, OR, NOT, IF-THEN).
- Propositional logic has limited expressiveness, as it cannot represent and reason about individual objects, their properties, and the relationships between them.

ACK: Prof. Endre Ózkan, UNIK of Tübingen, and Dr. Shao ZhiYuan, UNW

First Order (Predicate) Logic

- First-order logic (also known as predicate logic) is a more expressive and powerful formalism than propositional logic.
- It introduces the concepts of *individuals*, *predicates*, and *quantifiers* (universal and existential) to represent and reason about the world.
- *Quantifiers* (e.g., "for all," "there exists") allow for the expression of general statements and the reasoning about specific individuals.
- First-order logic can represent and reason about more complex and structured knowledge, such as hierarchies, relationships, and rules.



Example: FOL Sentence

- “Every rose has a thorn”

$$\forall X. (rose(X) \rightarrow \exists Y. (has(X, Y) \wedge thorn(Y)))$$

- For all X
 - if (X is a rose)
 - then there exists Y
 - (X has Y) and (Y is a thorn)



Example: FOL Sentence

- “On Mondays and Wednesdays, I go to John’s house for dinner”.

$$\forall X. ((is_mon(X) \vee is_wed(X)) \rightarrow \\ eat_meal(me, houseOf(john), X))$$

- Note the change from “and” to “or”
 - Translating is problematic

First Order Logic (Introduction)

Predicates are the statements involving variables which are neither **TRUE** nor **FALSE** if the values of the variables are not specified.

Example: 1) X is an animal
2) $a + b = 8$

<https://www.youtube.com/watch?v=ARywou8HLQk>



PREDICATES



Higher Order Logic (HOL)

- **Higher-Order Logic (HOL)** is an *extension* of First-Order Logic (FOL) where *functions, predicates, and even quantifiers can take other functions or predicates as arguments*. This makes HOL more expressive and powerful than FOL.
- Functions and predicates are also objects
 - Described by predicates: `binary(addition)`
 - Transformed by functions: `differentiate(square)`
 - Can quantify over both.
- E.g., define red functions as having zero at 17

$$\forall F. (red(F) \leftrightarrow F(0) = 17)$$

- Much harder to reason with.

ACK: Prof. Ender Özcan, UNUK, Prof. Tomas Maul & Dr Chen ZhiYuan, UNM



HOL vs FOL

Feature	First-Order Logic (FOL)	Higher-Order Logic (HOL)
Quantification	Only over objects (e.g., $\forall x, \exists x$)	Over objects, predicates, and functions (e.g., $\forall P, \exists f$)
Functions & Predicates	Cannot take predicates as arguments	Functions can take predicates/functions as arguments
Expressiveness	Limited expressiveness	More expressive (e.g., can define sets, relations, etc.)

ACK: Prof. Ender Özcan, UNUK, Dr Tomas Maul & Dr Chen ZhiYuan, UNM



Understanding Higher-Order Quantification

- In HOL, we can quantify over functions and predicates, not just objects.
- Example: "There exists a property that applies to all objects."

$$\exists P \forall x P(x)$$

- This means "There is some property that holds for all objects."



Defining Functions in HOL

- Example: A function f that doubles a number:
 - $\forall x (f(x)=2 \cdot x)$
- Example: Applying a function ff to every element of a set:
 - $\forall f \forall x (P(x) \rightarrow P(f(x)))$
 - This means: "For every function ff , if $P(x)$ holds, then $P(f(x))$ also holds."



Using Higher-Order Predicates

- Example: Universal Property of a Set
- $\forall S \forall P (\forall x (S(x) \rightarrow P(x)) \rightarrow \text{UniversalProperty}(S, P))$
 - $S(x)$ checks if x is in set S .
 - $P(x)$ is a property applied to elements of S .
- This states that if every element in S has property P , then S satisfies a universal property.



2
3

Expressing uniqueness

- Sometimes we want to say that there is a single, unique object that satisfies a certain condition
- “There exists a unique x such that $\text{king}(x)$ is true”
 - $\exists x \text{ king}(x) \wedge \forall y (\text{king}(y) \rightarrow x=y)$
 - $\exists x \text{ king}(x) \wedge \neg \exists y (\text{king}(y) \wedge x \neq y)$
 - $\exists! x \text{ king}(x)$
- “Every country has exactly one ruler”
 - $\forall c \text{ country}(c) \rightarrow \exists! r \text{ ruler}(c,r)$
- Iota operator: “ $\iota x P(x)$ ” means “the unique x such that $p(x)$ is true”
 - “The unique ruler of Freedonia is dead”
 - $\text{dead}(\iota x \text{ ruler}(\text{freedonia}, x))$

ACK: Prof. Ender Özç

RECORDED WITH

SCREENCAST



WANT TO

<https://www.youtube.com/watch?v=BflhowTu0jk>





Beyond True and False

- **Multi-valued logics**
 - More than two truth values.
 - e.g., **true, false & unknown**.
 - **Fuzzy logic** uses probabilities, truth value in $[0,1]$.
- **Modal logics** (John is happy $\rightarrow$ John is **usually** happy)
 - Modal operators define *mode* for propositions.
- **Epistemic logics** (belief)
 - e.g., $\Box p$ (necessarily p), $\Diamond p$ (possibly p), ...
 - “ **x knows that**”, “**For all x knows, it may be true that...**”
- **Temporal logics** (time)
 - e.g., $\Box p$ (always p), $\Diamond p$ (eventually p), ...
 - “ **P is the case at every time t** ”, “ **P is the case at some time t** ”,

ACK: Prof. Ender Özcan, UNUK, Prof. Tomas Maul & Dr Chen ZhiYuan, UNM



Logic is a Good Representation

- Fairly easy to do the translation when possible.
- Branches of mathematics devoted to it.
- It enables us to do logical reasoning
 - Tools and techniques come for free.
- Basis for programming languages
 - **Prolog** uses logic programs (a subset of FOL)
 - **λ Prolog** based on HOL

<https://en.wikipedia.org/wiki/Prolog>

Query Window

```
?- likes(john, jane). ← dot necessary
true. ← answer from prolog interpreter

sign on
prolog query
prompt
↓
?- friends(X, Y).
X = john,
Y = jane ; ← type ; to get next solution
X = jane,
Y = john.
```




Prolog

- **Prolog** is a *declarative* programming language designed for developing logic-based AI applications. Developers can set *rules* and *facts* around a problem, and then Prolog's interpreter will use that information to automatically infer solutions.
- <https://builtin.com/software-engineering-perspectives/prolog>
- <https://onecompiler.com/prolog>

Logic

Propositional

First Order Logic

Higher Order Logic





University of
Nottingham

UK | CHINA | MALAYSIA

Non-Logical Representations



Non-Logical Representations?

- Logic representations have restrictions / limitations and can be hard to work with.
 - Many AI researchers searched for better representations → *non-logical representations*
- **Production rules**
- **Semantic networks**
 - Conceptual graphs
 - Frames

ACK: Prof. Ender Özcan, UNUK, Prof. Tomas Maul & Dr Chen ZhiYuan, UNM



Production Rules



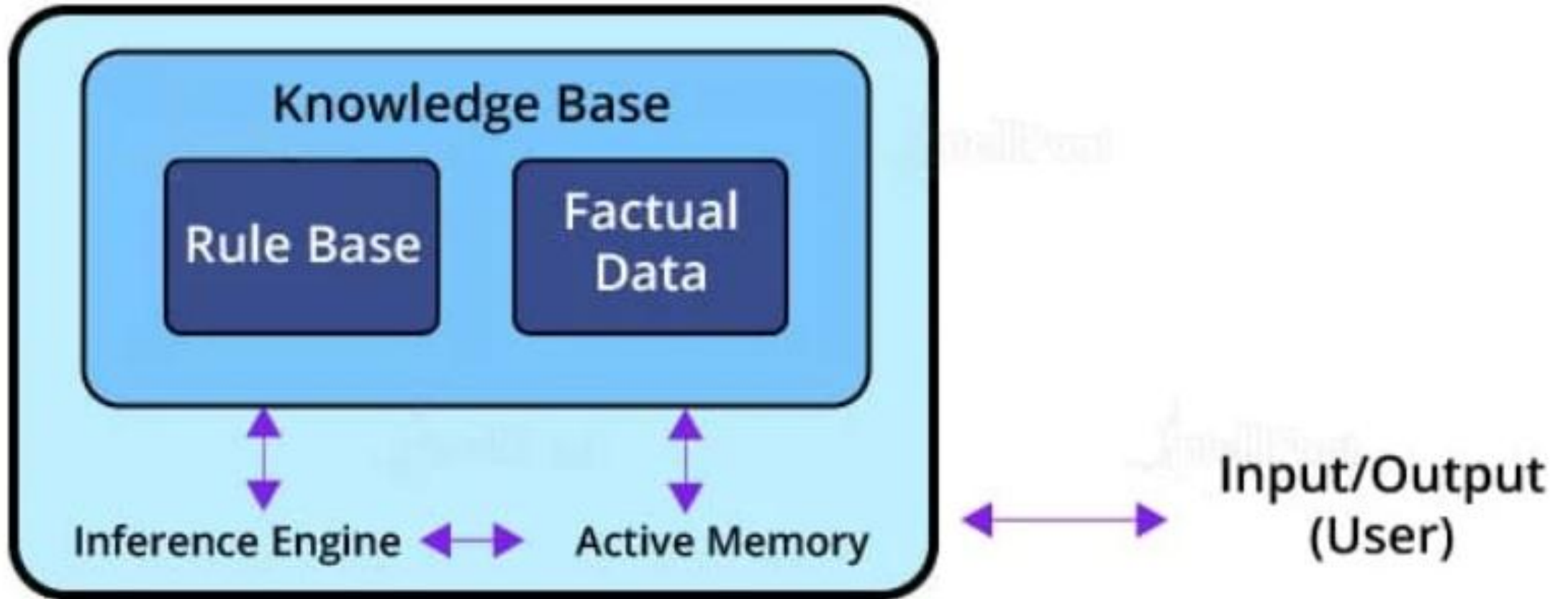
Production System

- A **production system** (or *production rule system*) is a computer program typically used to provide some form of artificial intelligence, which consists primarily of a set of **rules** about behavior, but also includes the mechanism necessary to follow those rules as the system responds to states of the world.
- Those **rules**, termed **productions**, are a basic knowledge representation found useful in automated planning and scheduling, expert systems, and action selection.

[https://en.wikipedia.org/wiki/Production_system_\(computer_science\)](https://en.wikipedia.org/wiki/Production_system_(computer_science))



Production Rule System



ACK: Prof. Ender Özcan, UNUK, Dr Tomas Maul & Dr Chen ZhiYuan, UNM



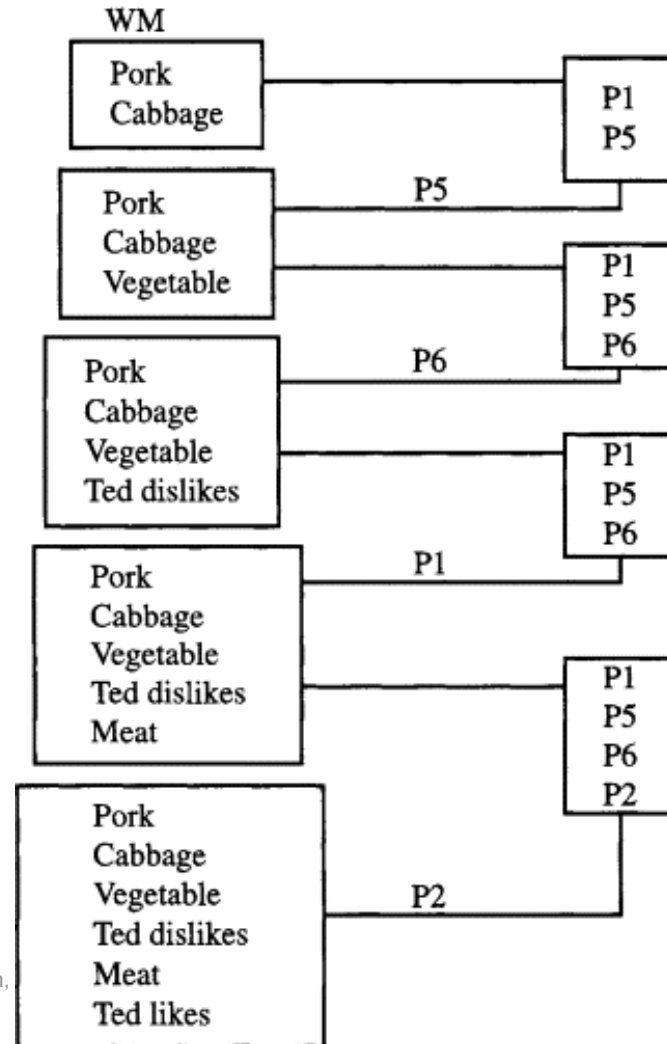
Production Rule

- **Production rules**, also known as *condition-action rules* or IF-THEN rules, are a fundamental concept in *rule-based systems* and *expert systems*. They are a way of representing and reasoning about knowledge in a *declarative form*.
- Production rules for knowledge representation:
 - Their “modularity” (each **rule** represents one “unit” of knowledge, simplifying knowledge acquisition and maintenance of the knowledge base), and
 - The ability of domain experts to articulate their knowledge in the form of rules.



- P1 IF Pork THEN Meat
P2 IF Meat THEN Ted likes
P3 IF Chocolate THEN Cookie
P4 IF Cookie THEN Hannah likes
P5 IF Cabbage THEN Vegetable
P6 IF Vegetable THEN Ted dislikes

Production Rules





A production rule consists of two parts:

1. The condition (IF-part):

- This part specifies the set of *conditions* or *premises* that must be satisfied for the rule to be applicable.
- The conditions are typically expressed as logical expressions or patterns that must match the current *state* of the system.

2. The action (THEN-part):

- This part specifies the *actions* or *consequences* that should be taken if the conditions in the IF-part are met.
- The actions can involve modifying the system's state, triggering further processing, or producing some output.



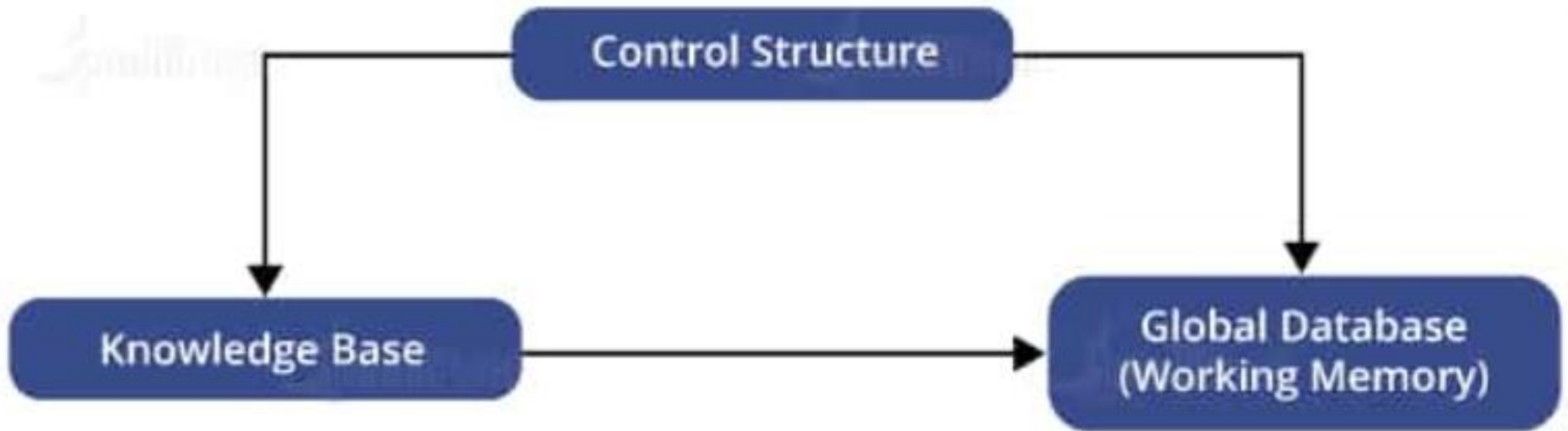
Production Rules

- **Rule set** of <condition,action> pairs
 - “**IF** condition **THEN** action”
- **Match-resolve-act** cycle
 1. **Match (Recognize)**: Agent checks if each rule’s condition holds (based on state of the system).
 2. **Resolve**:
 - Multiple production rules may fire at once (**conflict set**)
 - Agent must choose rule from set (**conflict resolution**)
 3. **Act**: If so, rule “fires” and the action is carried out.
- **Working memory**:
 - rule can write knowledge to working memory.
 - knowledge may match and fire other rules.

ACK: Prof. Ender Özcan, Yildiz Teknik Üniversitesi, Ziraat Fakültesi, Ziraat Mühendisliği Bölümü



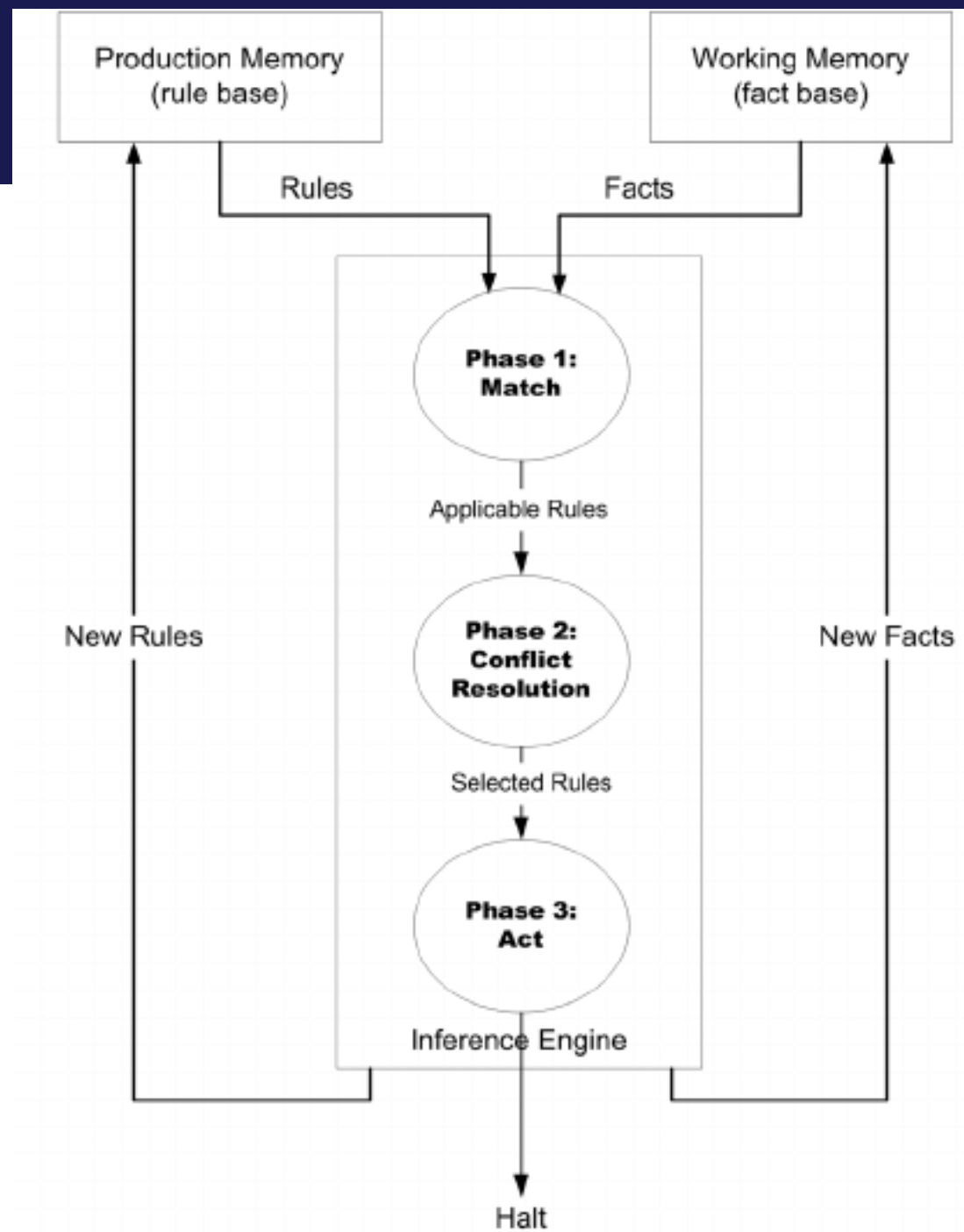
Production Rule System



ACK: Prof. Ender Özcan, UNUK, Dr Tomas Maul & Dr Chen ZhiYuan, UNM



Example





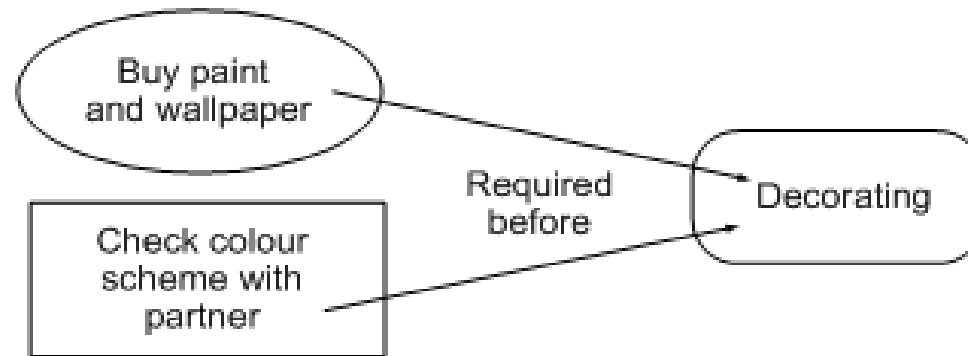
Semantic networks

Conceptual graphs
Frames

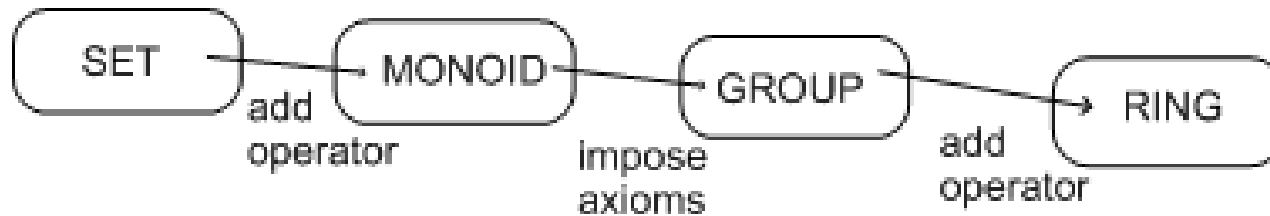


Graphical Representation

- Humans draw diagrams all the time, e.g.
 - Causal relationships



- And relationships between ideas

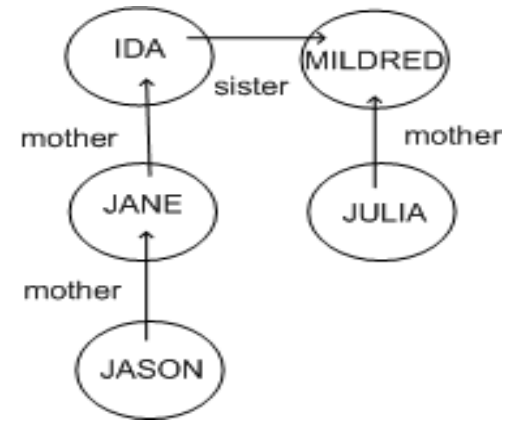
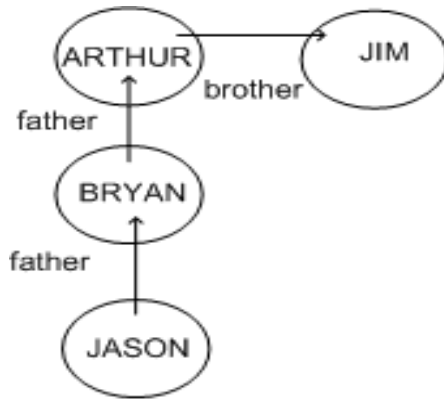


ACK: Prof. Ender Özcan, UNUK, Prof. Tomas Maul & Dr Chen ZhiYuan, UNM



Graphical Representation

- **Graphs** easy to store in a computer.
- To be of any use must impose a formalism.



- Jason is 15, Bryan is 40, Arthur is 70, Jim is 74
- How old is Julia?



Semantic Networks (SN)

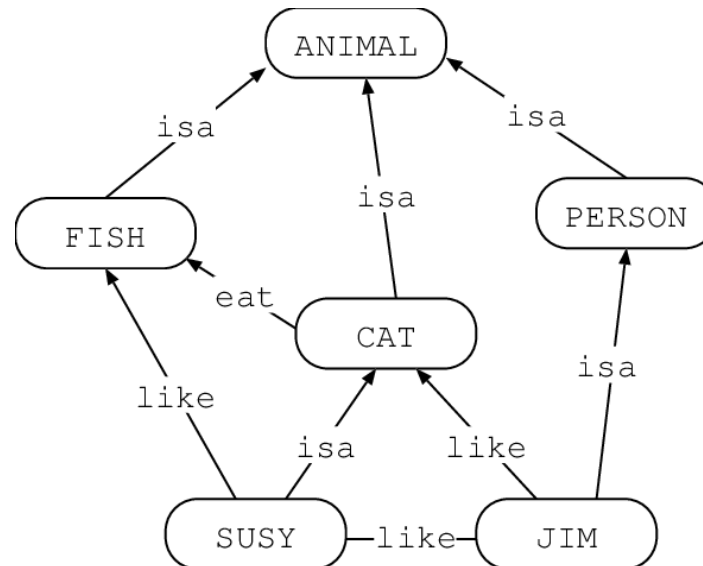
- Graphical representation (a **graph**)
 - Links indicate subset, member, relation, ...
- Equivalent to logical statements (usually FOL)
 - Easier to understand than FOL?
 - Specialised Semantic Networks reasoning algorithms can be faster.
- Example: natural language understanding
 - Represent the meaning of sentences using a structured network of concepts.
 - Focus on concepts instead of syntax
 - Sentences with the same meaning have same graphs.

-----Conceptual Dependency Theory (Schank)
https://en.wikipedia.org/wiki/Conceptual_dependency_theory



Semantic Network

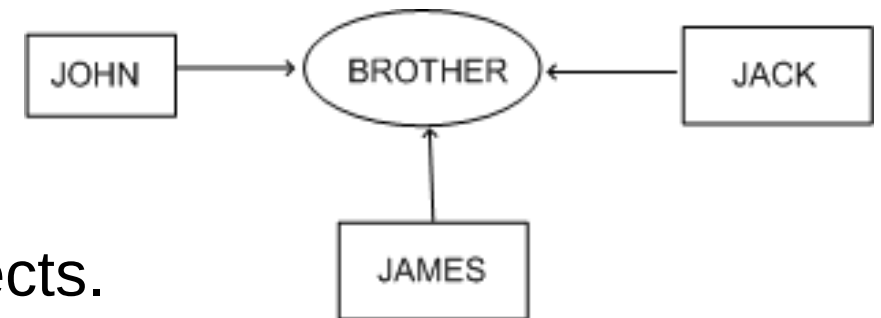
- A **semantic network** is a knowledge base that represents semantic relations between concepts in a network.
- It is a directed or undirected graph consisting of vertices / nodes, which represent **concepts** or **entities**, and edges, which represent **semantic relations** between concepts (e.g., "is-a", "has-part", "causes"), mapping or connecting semantic fields.





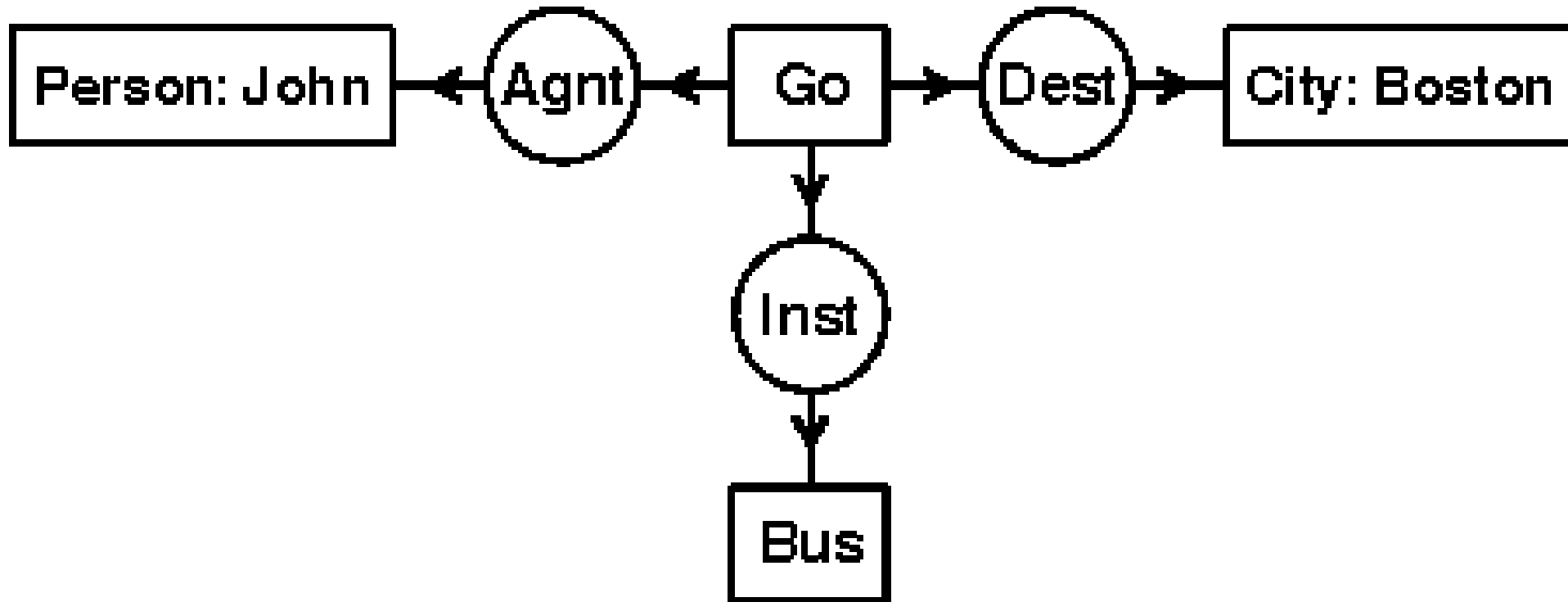
Conceptual Graphs

- **Semantic network** where each graph represents a single proposition, that follows formal logic (usually based on FOL). Introduced by John F. Sowa in the 1980s for natural language understanding and AI reasoning.
- The conceptual graph in represents a typed or sorted version of logic.
- **Concept nodes** can be
 - *Concrete* (visualisable) such as restaurant, my dog Spot.
 - *Abstract* (not easily visualisable) such as anger.
- **Edges** do not have labels.
 - Instead, conceptual relation nodes.
 - Easy to represent relations between multiple objects.





Example: Conceptual Graph



John is going to Boston by bus.



Example: Conceptual Graph

- Each of the four concepts has a type label, which represents the type of entity the concept refers to: Person, Go, Boston, or Bus.
- Two of the concepts have names, which identify the referent: John or Boston. Each of the three conceptual relations has a type label that represents the type of relation: agent (Agnt), destination (Dest), or instrument (Inst).
- The CG as a whole indicates that the person John is the agent of some instance of going, the city Boston is the destination, and a bus is the instrument.
- Can be translated to the following formula:
 - $(\exists x)(\exists y)(\text{Go}(x) \wedge \text{Person}(\text{John}) \wedge \text{City}(\text{Boston}) \wedge \text{Bus}(y) \wedge \text{Agnt}(x, \text{John}) \wedge \text{Dest}(x, \text{Boston}) \wedge \text{Inst}(x, y))$

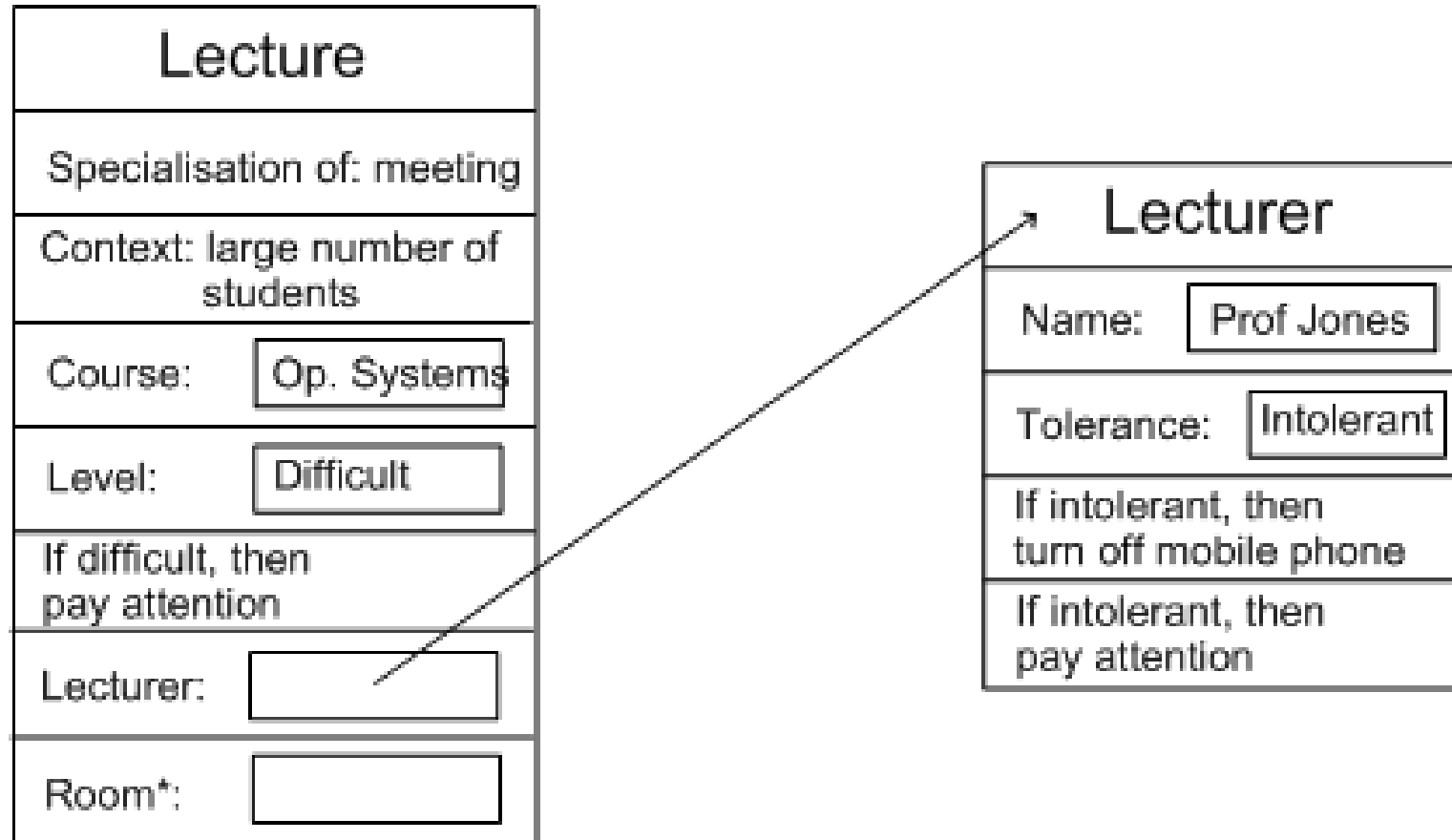


Frame Representations

- Semantic networks where nodes have structure → adding structured **attributes (slots)** and **values**.
 - **Frame** with a number of slots (age, height, ...)
 - Each **slot** stores specific item of information with value.
- When agent faces a new situation
 - Slots can be filled in (value may be another frame)
 - Filling in may trigger actions
 - May trigger retrieval of other frames
- *Inheritance* of properties between frames.
 - useful for representing object-oriented and event-based



Example: Frame Representation



ACK: Prof. Ender Özcan, UNUK, Prof. Tomas Maul & Dr Chen ZhiYuan, UNM



The main components of a frame representation are:

1. Frames:

- Frames represent stereotypical *situations, objects, or concepts*.
- Frames are typically represented as data structures or objects with slots.

2. Slots:

- Slots are the individual *components or attributes* of a frame.
- Slots can contain various types of information, such as *values, default values, constraints, or procedures*.

3. Fillers (actual values assigned to those slots):

- Fillers are the specific values or data that are assigned to the slots in a frame.

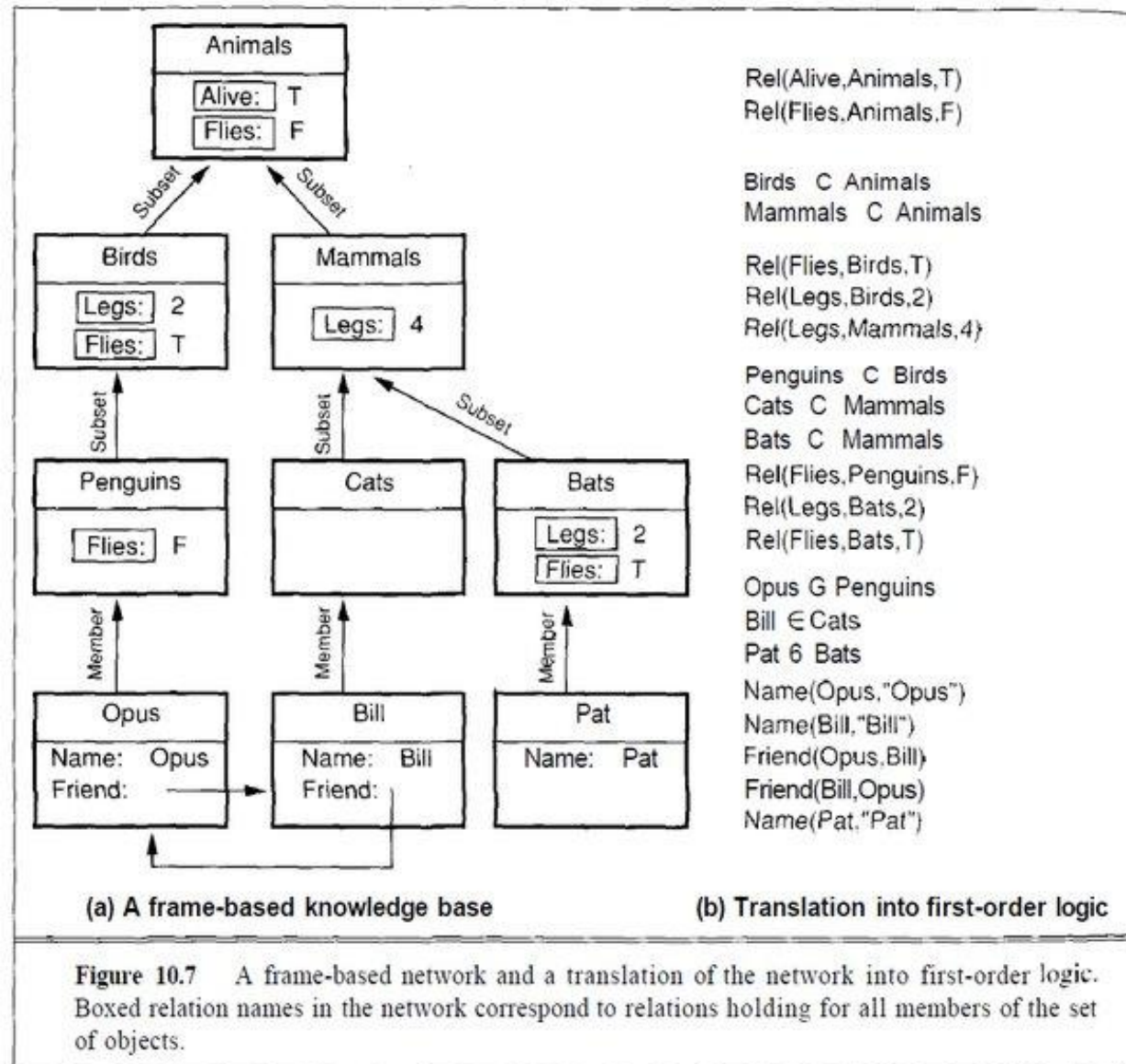


Flexibility in Frames

- **Slots** in a frame can contain:
 - *Information* for choosing a frame in a situation.
 - *Relationships* between this and other frames.
 - *Procedures* to carry out after various slots filled.
 - *Default information* to use where input is missing.
 - *Blank* slots: left blank unless required for a task.
 - Other frames, which gives a hierarchy.
- Can also be expressed in first order logic (FOL).



Example: Frame Representation





Representation & Logic

- Non-logical representations
 - Production rules
 - Semantic networks
 - Conceptual graphs, frames
- But all can be expressed in **first order logic!**
- Best of both worlds.
 - Logical reading ensures representation well-defined.
 - Representations specialised for applications.
 - Can make reasoning easier, more intuitive.

ACK: Prof. Ender Özcan, UNUK, Prof. Tomas Maul & Dr Chen ZhiYuan, UNM



University of
Nottingham

UK | CHINA | MALAYSIA

Case-based Reasoning



Reasoning

- What is **reasoning**? In general, it is the *formal manipulation of the symbols* representing a collection of believed propositions to *produce representations of new ones* → understanding the meaning of the representation (symbols)
- Here that we use the fact that symbols are more accessible than the propositions they represent:
 - They must be concrete enough that we can manipulate them (move them around, take them apart, copy them, string them together) in such a way as to construct representations of new propositions.
 - We might start with the sentences “**John loves Mary**” and “**Mary is coming to the party**” and after a certain amount of manipulation produce the sentence, “**Someone John loves is coming to the party**”.
 - We would call this form of reasoning “**logical inference**” because the final sentence represents a logical conclusion of the propositions represented by the initial ones.



Types of Reasoning

- **Inductive reasoning:** This involves observing specific instances or observations and then generalizing to broader conclusions or theories. Inductive reasoning is based on probabilities rather than certainties.
- **Deductive reasoning:** This involves starting with general principles or premises and logically deriving specific conclusions from them. The conclusions necessarily follow from the premises.



Types of Reasoning

Aspect	Inductive Reasoning	Deductive Reasoning
Definition	A reasoning process where generalizations are formed based on specific observations or evidence.	A reasoning process that starts with a general premise and derives specific conclusions.
Direction	Specific → General	General → Specific
Basis	Observation, patterns, or evidence.	Established facts, rules, or premises.
Nature of Conclusion	Probabilistic: The conclusion may be likely but not guaranteed to be true.	Certain: The conclusion must be true if the premises are true.
Example	Observing that the sun rises every day → Concluding that the sun will rise tomorrow.	Premise: All humans are mortal. Premise: Socrates is human. → Conclusion: Socrates is mortal.

ACK: Prof. Ender Ozcan, UNUK, Dr Tomas Maul & Dr Chen ZhiYuan, UNM



Types of Reasoning

- **Abductive reasoning:** This involves starting with an observation and then developing the most likely explanation or hypothesis to account for that observation. Abductive reasoning is often used in scientific inquiry. It is often described as "inference to the best explanation."
- **Analogical reasoning:** This involves using similarities between known cases or situations to draw conclusions about unknown cases or situations → **case-based reasoning**



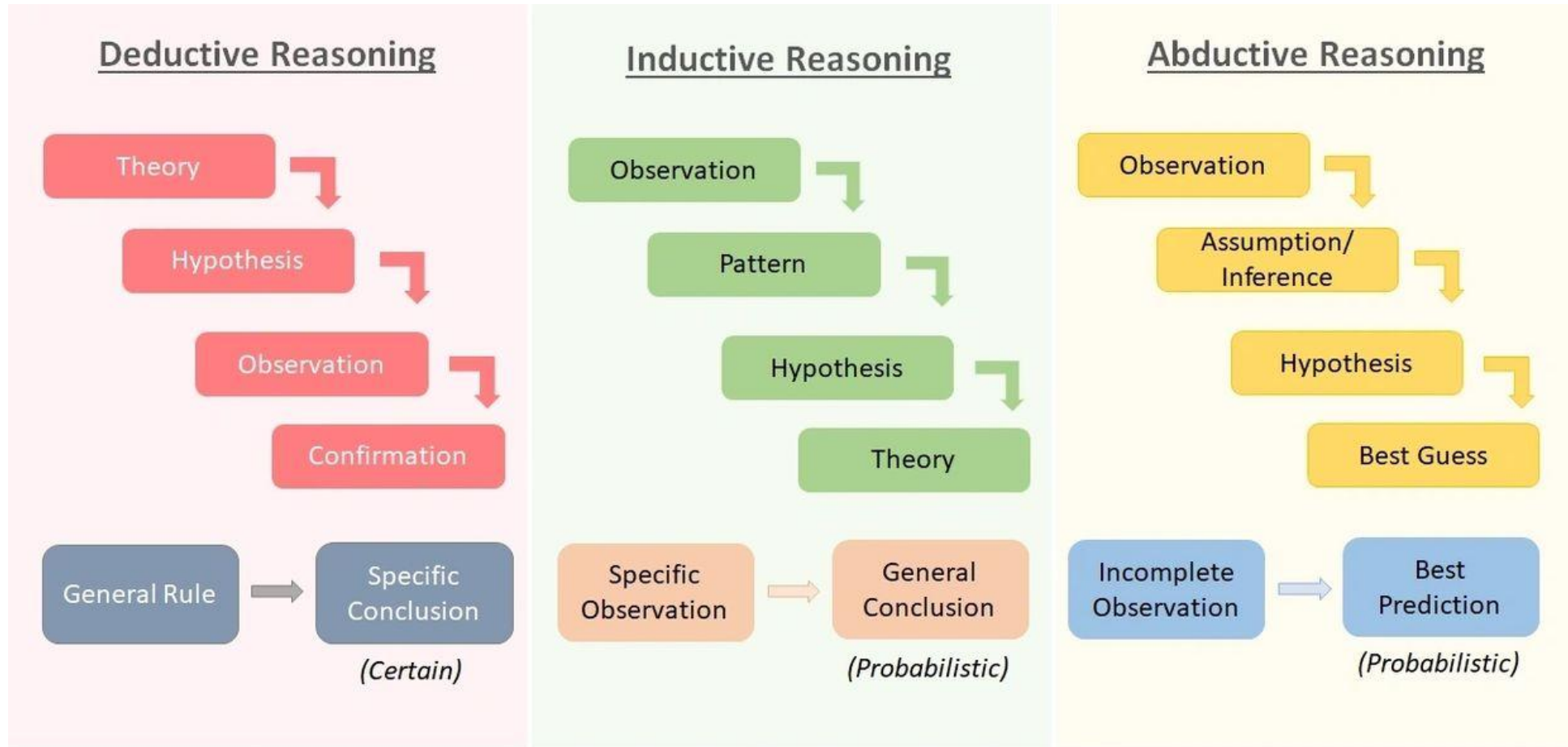
Types of Reasoning

 Inductive Reasoning	Starts with specific observations, then draws general conclusions.	You see your dog wags its tail every time you get home. You conclude that seeing you makes your dog happy (might not always be true!).	Like a detective gathering clues to solve a mystery.
 Deductive Reasoning	Starts with a general rule (hypothesis) and applies it to specific cases.	You know all birds have feathers (general rule). You see a robin (specific case). You conclude the robin has feathers.	Like applying a recipe (general instructions) to bake a specific cake.
 Abductive Reasoning	Starts with an observation and proposes a possible explanation, often the most likely one.	You hear a loud crash in the kitchen. You find a broken vase on the floor. You conclude someone might have dropped it accidentally.	Like a doctor making a diagnosis based on symptoms.

ACK: Prof. Ender Özcan, UNUK, Dr Tomas Maul & Dr Chen ZhiYuan, UNM



Types of Reasoning



ACK: Prof. Ender Özcan, UNUK, Dr Tomas Maul & Dr Chen ZhiYuan, UNM



Case-Based Reasoning

- **Case-Based reasoning (CBR)**, is the process of solving new problems based on the solutions of similar past problems (cases).
- CBR is reasoning by *remembering*: It is a starting point for new reasoning.
- CBR is a well-established research field that involves the investigation of theoretical foundations, system development and practical application building of experience-based problem solving.



Case-Based Reasoning

Everyday examples of CBR :

- An *auto mechanic* who fixes an engine by recalling another car that exhibited similar symptoms.
- A *lawyer* who advocates a particular outcome in a trial based on legal precedents or a judge who creates case law.
- An *engineer* copying working elements of nature (practicing biomimicry), is treating nature as a database of solutions to problems.

*Case-based reasoning is a prominent kind of **analogy making**.



Key aspects of case-based reasoning

1. **Storing** previous cases or experiences
2. **Retrieving** relevant cases
3. **Adapting** the retrieved solution
4. **Evaluating** the adapted solution
5. **Learning** and **updating** the knowledge base

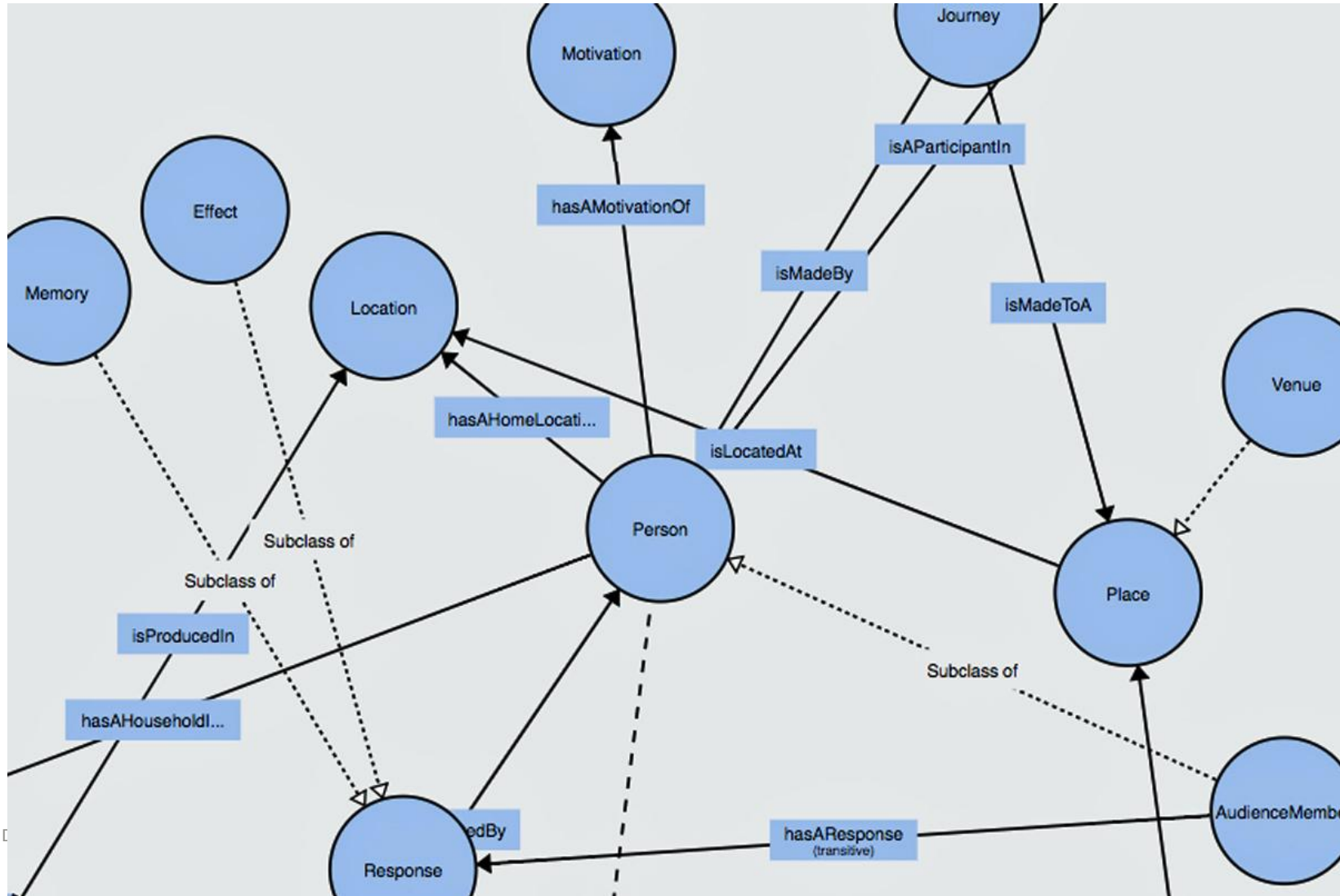


Types of CBR

- There are three main types of CBR that differ significantly from one another concerning case representation and reasoning:
 1. **Structural** (a common *structured vocabulary* (collection of attributes, properties, and relationships), i.e., an ontology)
 2. **Textual** (cases are represented as *free text*, i.e., strings)
 3. **Conversational** (a case is represented through a list of *questions* that varies from one case to another; knowledge is contained in customer / agent *conversations* (Q&A))



Ontology



ACK: Prof. Ender Özcan, UNUK, I



Types of CBR

Type	Representation	Reasoning Method	Example Use Case
Structural	Structured attributes (ontology)	Attribute-based similarity	Medical diagnosis, technical troubleshooting
Textual	Free-text documents (strings)	NLP, keyword matching, embeddings	Legal case search, document retrieval
Conversational	Q&A-based (dialogue)	Interactive question refinement	Customer support, troubleshooting

ACK: Prof. Ender Özcan, UNUK, Dr Tomas Maul & Dr Chen ZhiYuan, UNM



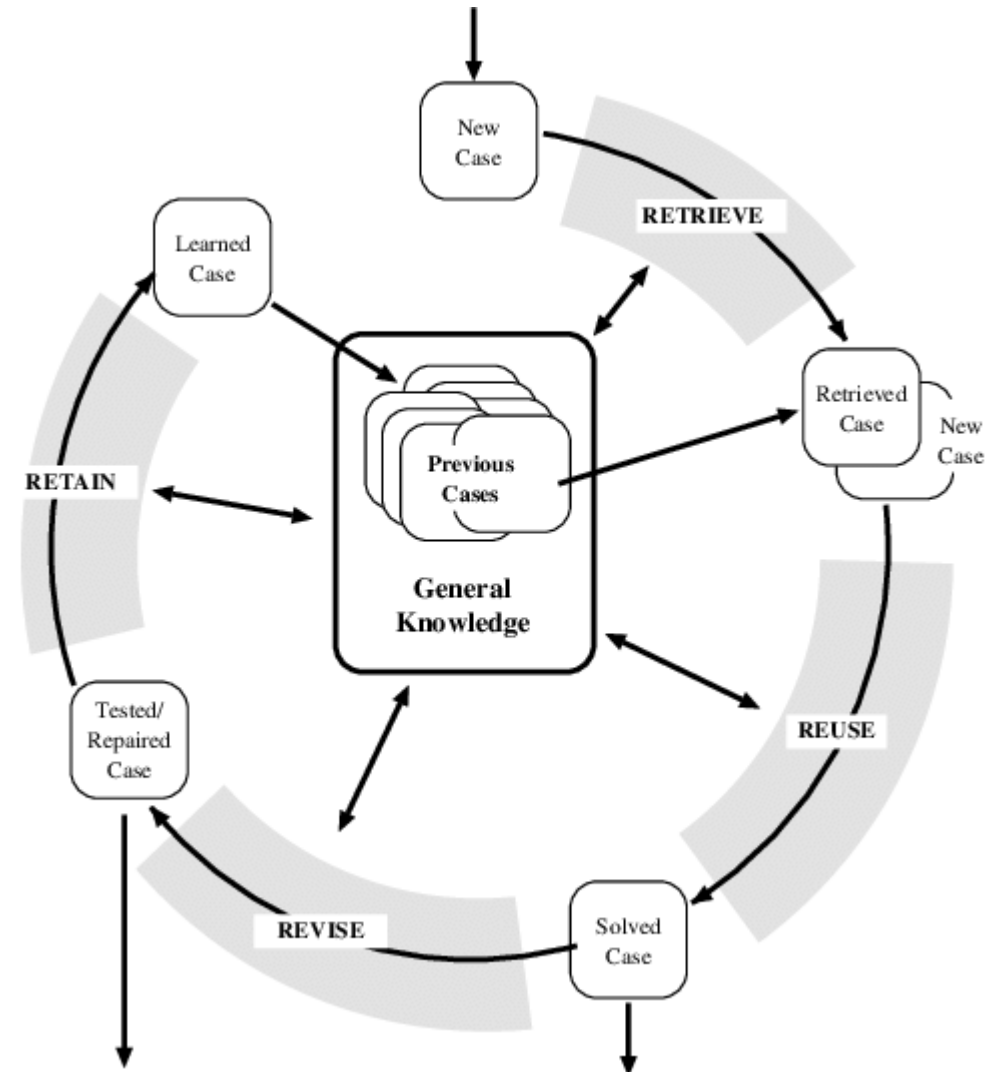
Types of CBR

Aspect	Structural	Textual	Conversational
Flexibility	Low	High	High
Retrieval	Efficient (structured queries)	Moderate (NLP-based)	Complex (sequence and context-dependent)
Richness of Data	Limited to predefined schema	High (free text allows detailed capture)	High (captures problem evolution)
Scalability	High (with indexing)	Moderate (text requires NLP processing)	Low (due to sequential nature)
Complexity	Moderate (requires schema design)	High (requires NLP models)	Very High (requires dialog understanding)
Best Fit Domains	Medical, legal, technical troubleshooting	Customer support, incident management	Customer service, help desk interactions



CBR Cycle

- The **CBR cycle** is proposed by Aamodt and Plaza.
- The CBR cycle consists of 4 sequential steps (4R) around the knowledge of the CBR system.



ACK: Prof. Ender Özcan, UNUK, Dr Tomas Maul & Dr Chen ZhiYuan, UNM



1. Retrieve

- The retrieval task is defined as finding a small number of cases from the case-base with the *highest similarity to the query*.
- This could be a *k-nearest-neighbor retrieval* task considering a specific similarity function.
- When the case base grows, the efficiency of retrieval decreases => methods that improve retrieval efficiency, e.g., specific index structures such as kd-trees, case-retrieval nets, or discrimination networks.



2. Reuse

- Reusing a retrieved solution can be quite simple if the solution is returned unchanged as the proposed solution for the new problem (without adaptation).
- *Adaptation* (if required).
- Several techniques for adaptation in CBR
 - **Transformational** adaptation
 - **Generative** adaptation
- Most practical CBR applications today try to avoid extensive adaptation for pragmatic reasons.

ACK: Prof. Ender Özcan, UNUK, Prof. Tomas Maul & Dr Chen ZhiYuan, UNM



3. Revise

- In this phase, *feedback* related to the solution constructed so far is obtained.
- This feedback can be given in the form of a correctness rating of the result or in the form of a manually corrected revised case.
- The *revised case* or any other form of feedback enters the CBR system for its use in the subsequent retain phase.



4. Retain

- The retain phase is the *learning* phase of a CBR system (adding a revised case to the case base).
- Explicit *competence models* have been developed that enable the *selective retention* of cases (because of the continuous increase of the case-base).
- The revised case or any other form of feedback enters the CBR system for its use in the subsequent retain phase.



CBR applications

- During the past 20 years, many CBR applications have been developed, ranging from prototypical applications build in research labs to large-scale fielded applications developed by commercial companies.
- **Application areas** of CBR include:
 - help-desk and customer service.
 - recommender systems in electronic commerce.
 - knowledge and experience management.
 - medical applications and applications in image processing.
 - applications in law, technical diagnosis, design, planning.
 - applications in the computer games and music domain.

ACK: Prof. Ender Özcan, UNUK, Prof. Tomas Maul & Dr Chen ZhiYuan, UNM



Case-Based Reasoning (CBR) vs. Transfer Learning

- Both CBR and Transfer Learning aim to **leverage past knowledge** to solve new problems more effectively. They share the idea of reusing prior experiences, but they do so in different ways:
 1. **CBR** focuses on retrieving and adapting specific past cases to solve new problems. It is widely used in expert systems, recommendation systems, and decision support.
 2. **Transfer Learning** focuses on **reusing** learned representations from one domain or task to improve performance on another, often different, but related task.
- **Analogy:**
 - CBR is like a doctor who recalls past patient cases to diagnose and treat a new patient.
 - Transfer Learning is like an experienced doctor who has trained in one specialty (e.g., cardiology) but can apply their foundational knowledge to another specialty (e.g., neurology) with some adjustments.



Case-Based Reasoning (CBR) vs. Transfer Learning

Feature	Case-Based Reasoning (CBR)	Transfer Learning
Knowledge Representation	Stores and retrieves past cases (explicit memory)	Learns generalizable features from data (implicit memory)
Learning Mechanism	Uses similarity measures to retrieve and adapt past solutions	Fine-tunes pre-trained models or reuses learned features
Scope of Application	Decision-making, expert systems, problem-solving	Deep learning, neural networks, computer vision, NLP
Adaptation	Requires human-defined or algorithmic case adaptation	Adapts learned weights via fine-tuning

ACK: Prof. Ender Özcan, UNUK, Dr Tomas Maul & Dr Chen ZhiYuan, UNM



University of
Nottingham

UK | CHINA | MALAYSIA

Decision Making, Decision Support



Definition – decision

- A **decision** is defined as the choice of one among a number of alternatives.
- "**Decision Making** is the process of *choosing among alternative courses* of action for the purpose of attaining a goal or goals.“ - Turban and Aronson, 1998
- "Every decision-making process produces a *final choice*. The output can be an action or an opinion of choice.“ - Wikipedia



Decision Making

Decision making is the whole process of making the choice:

1. *assessing* the problem,
2. *collecting* and *verifying* information,
3. *identifying* alternatives,
4. *anticipating* consequences of decisions,
5. *making the choice* using sound and logical judgement based on available information,
6. *informing* others of decision and rationale,
7. *evaluating* decisions.

Source: <http://kt.ijs.si/MarkoBohanec/pub/WhatDS.pdf>

ACK: Prof. Ender Özcan, UNUK, Prof. Tomas Maul & Dr Chen ZhiYuan, UNM



Decision Making - Why?

- Increasingly complicated than in the past.
 - Available *alternatives* of technologies.
 - *Increased cost* due to complexity and magnitude of operations.
 - *Huge amount* of information, sometimes difficult to access.
 - *Quick* decisions.

- Managers need tools and techniques to support effective decision making.
 - *Computerised support* extensively used in modern business and management.
 - Generating *reports* → support making important decisions.



Decision Support

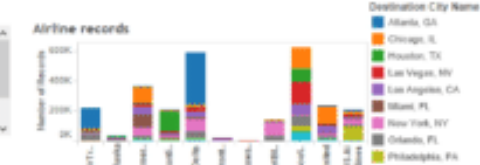
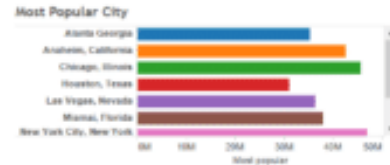
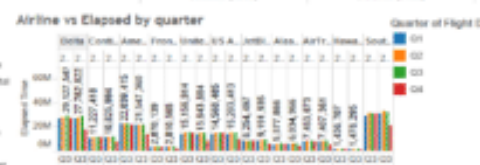
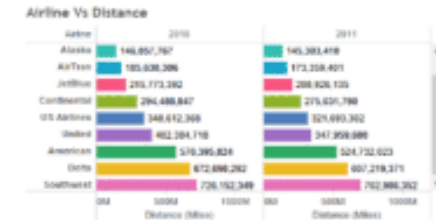
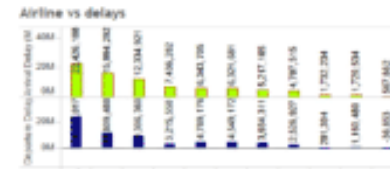
- This term is used often and in a variety of contexts related to decision making.
- Our focus will be:
 - utilizing computer-based systems that facilitate the use of *data*, *models*, and *structured decision processes* in decision making/solving optimisation problems for decision support.
- **Multidisciplinary:** Decision Theory, Decision Analysis, Operations Research, Statistics, Management Science, and Artificial Intelligence...

ACK: Prof. Ender Özcan, UNUK, Prof. Tomas Maul & Dr Chen ZhiYuan, UNM



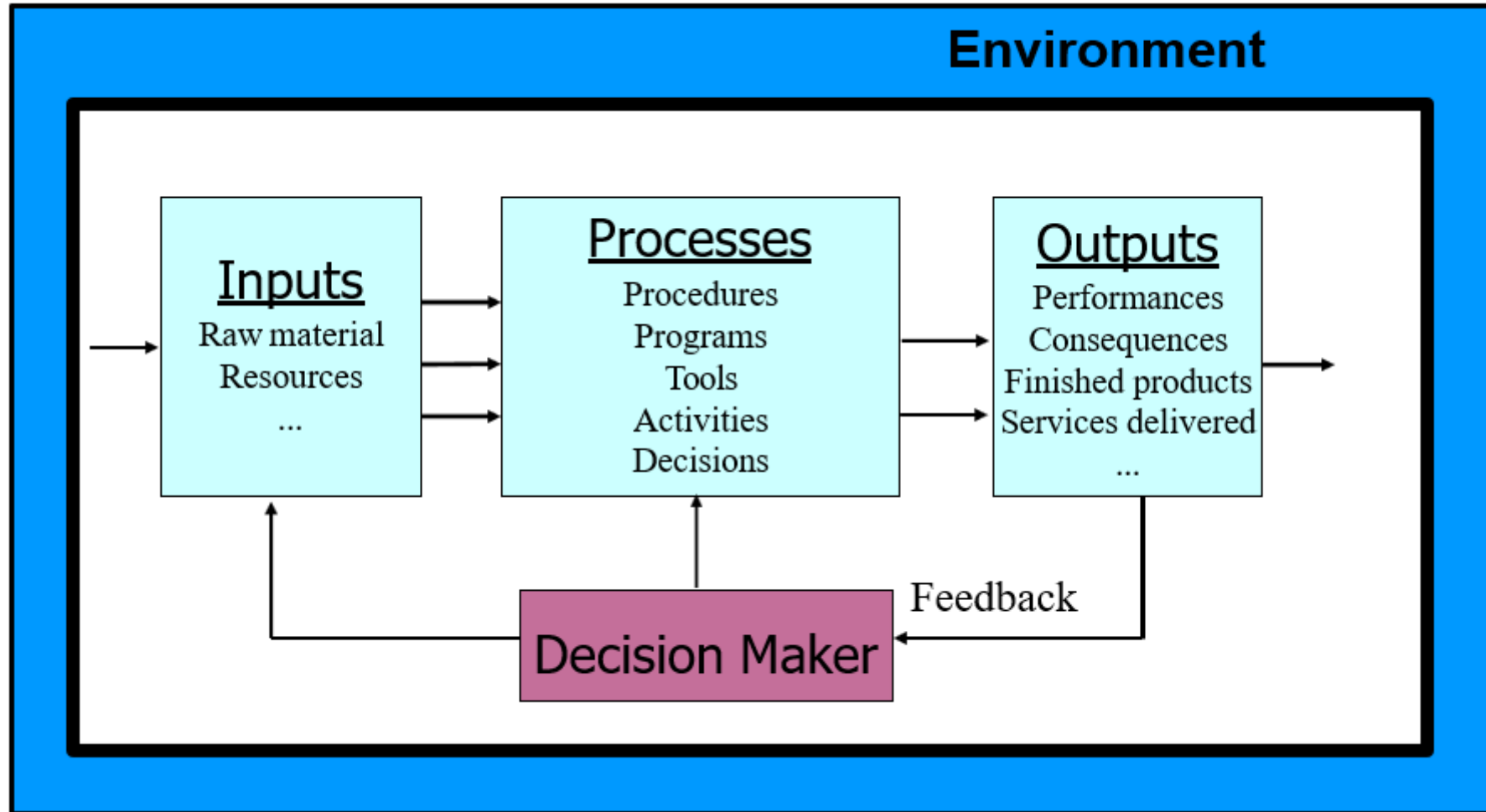
Decision Support Systems: Visual Perception and Data Visualization

Country	Postal Code	Market	Region	Product ID	Category	Sub-Category	Product Name	Sales
United States	10024	US	East	TEC-AC-1000	Technology	Accessories	Plantronics C	2309.65
Australia		APAC	Oceania	FUR-CH-1000	Furniture	Chairs	Novimex Exe	3709.395
Australia		APAC	Oceania	TEC-PH-1000	Technology	Phones	Nokia Smart	5175.171
Germany		EU	Central	TEC-PH-1000	Technology	Phones	Motorola Sm	2892.51
Senegal		Africa	Africa	TEC-SHA-100	Technology	Copiers	Sharp Wirele	2832.96
Australia		APAC	Oceania	TEC-PH-1000	Technology	Phones	Samsung Sm	2862.675
New Zealand		APAC	Oceania	FUR-CH-1000	Furniture	Chairs	Novimex Exe	1822.08
New Zealand		APAC	Oceania	FUR-TA-1000	Furniture	Tables	Chromcraft C	5244.84
United States	95823	US	West	OFF-BI-1000	Office Suppli	Binders	Fellowes PB5	5083.96
United States	28027	US	South	FUR-TA-1000	Furniture	Tables	Chromcraft B	4297.644
United States	22304	US	South	OFF-SU-1000	Office Suppli	Supplies	Martin Yale C	4164.05
Afghanistan		APAC	Central Asia	FUR-TA-1000	Furniture	Tables	Bevis Confer	4626.15
Saudi Arabia		EMEA	EMEA	TEC-CIS-1000	Technology	Phones	Cisco Smart f	2616.96
Brazil		LATAM	South	FUR-CH-1000	Furniture	Chairs	Harbour Crea	2221.8
China		APAC	North Asia	OFF-AP-1000	Office Suppli	Appliances	KitchenAid M	3701.52
France		EU	Central	OFF-AP-1000	Office Suppli	Appliances	Breville Refri	1869.588
United States	42420	US	South	TEC-AC-1000	Technology	Accessories	Logitech dNi	2249.91
Italy		EU	South	OFF-AP-1000	Office Suppli	Appliances	Hoover Stove	7958.58
Australia		APAC	Oceania	TEC-CO-1000	Technology	Copiers	Brother Fax M	2565.594
Tanzania		Africa	Africa	OFF-KIT-1000	Office Suppli	Appliances	KitchenAid St	3409.74
Poland		EMEA	EMEA	FUR-HON-10	Furniture	Tables	Hon Comput	1977.72
United States	60610	US	Central	TEC-PH-1000	Technology	Phones	Apple iPhone	2735.952
China		APAC	North Asia	FUR-CH-1000	Furniture	Chairs	SAFCO Execu	2754





Definition - System





Definition - System

- **System:** a collection of objects such as people, resources, concepts, and procedures intended to perform a function or to serve a goal.
- Turban and Aronson, 1998
- **Input:** elements that enter the system.
- **Processes:** convert or transform the inputs into outputs.
- **Output:** describe the finished products or the consequences of being in the system.
- **Feedback:** flow of information from the output concerning the system's performance; the decision maker may modify the inputs or processes.
- **Environment:** the elements that lie outside but impact the system's performance.

ACK: Prof. Ender Özcan, UNUK, Prof. Tomas Maul & Dr Chen ZhiYuan, UNM



Definition - System

- **Degree of independence** of systems.
 - **Closed** systems are totally independent.
 - **Open** systems dependent on their environment.
- **Evaluations** of systems
 - **System effectiveness**: the degree to which *goals are achieved*, i.e., result, output.
 - **System efficiency**: a measure of the use of inputs (or resources including time) to achieve outputs.



Definition - Models

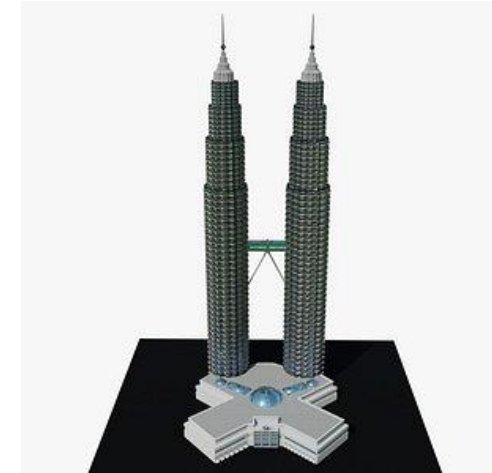
- Major characteristic of **Decision Support System (DSS)** is that it includes at least one **model**.
- A **model** is a simplified representation or abstraction of reality.
 - Reality is usually too *complex*.
 - Much of *complexity* is irrelevant to solving the problem.



Types of Models (based on representation)

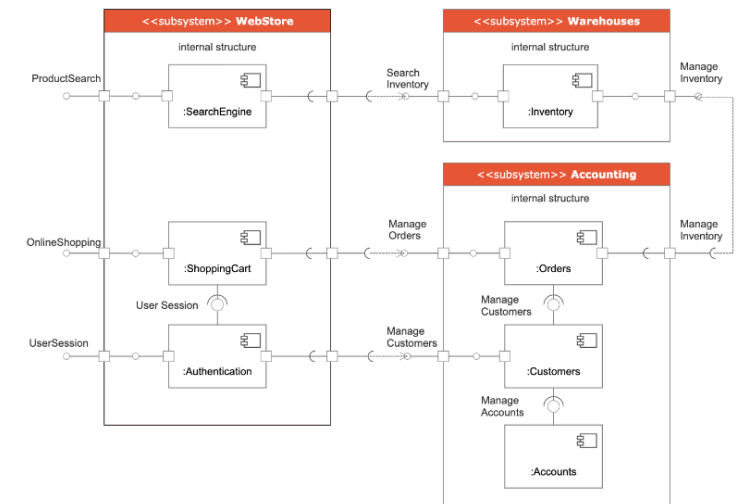
■ Iconic Models (physical)

- Least abstract - a physical replica of a system.
- 3-D, i.e. airplane, car
- 2-D, i.e. photos, GUI



■ Analog models (logical)

- More abstract - a *symbolic representation* of reality.
- Behaves like the real system but does not look like it.
- 2-D charts, diagrams, graphs



ACK: Prof. Ender Özcan, UNUK, Prof. Tomas Maul & Dr Chen ZhiYuan, UNM

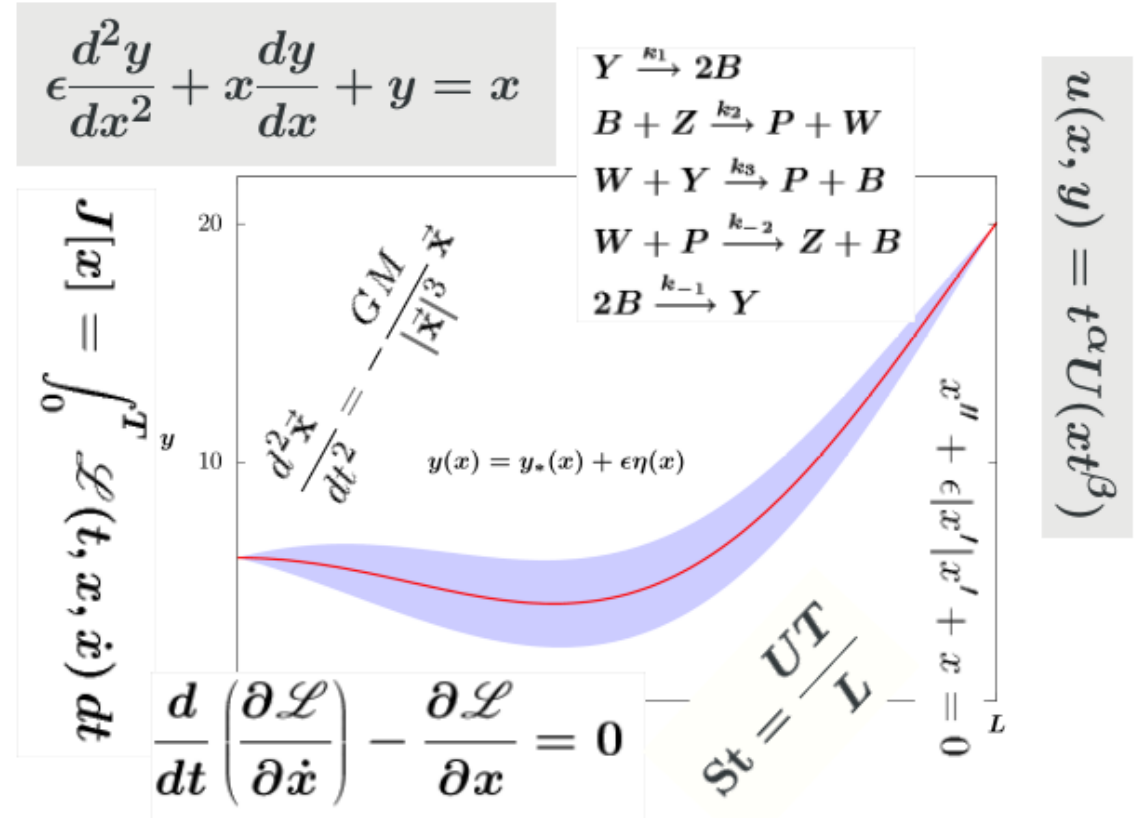


Types of Models

■ Mathematical Models

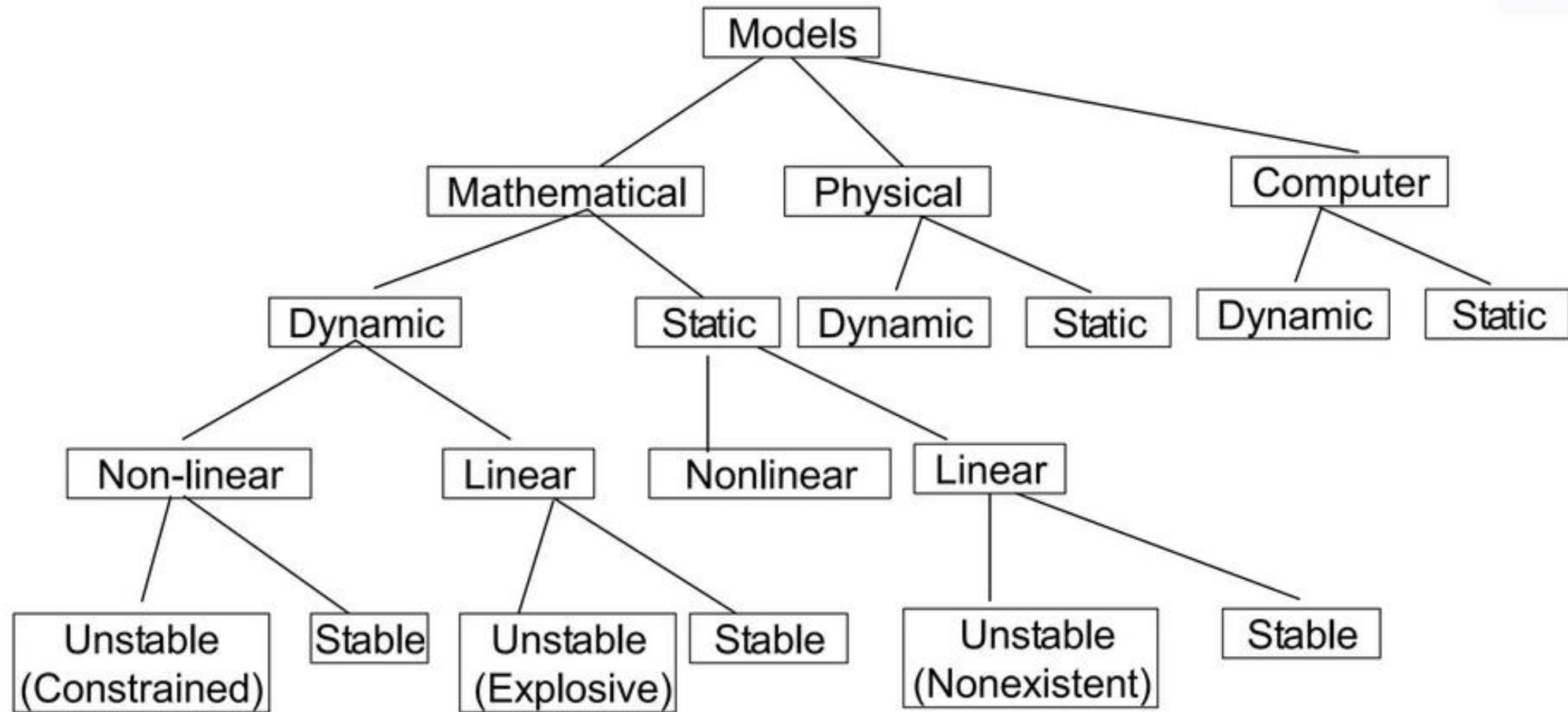
- Most *abstract* - mathematical relationships in the problem.
- Complexity of relationships is difficult to be represented by icons.
- Easy to model than manipulating real system.
- Low cost of the analysis.
- Cost of making mistakes is less than mistakes on real system.
- Can model risk and uncertainty.
- A very large number of solutions can be analysed.

ACK: Prof. Ender Özcan, UNUK, Prof. Tomas Maul & Dr Chen ZhiYuan, UNM





Classifications of Models



ACK: Prof. Ender Özcan, UNUK, Dr Tomas Maul & Dr Chen ZhiYuan, UNM



Example – The Problem

- A company (hospital, supermarket, etc)
 - A limited number of workers
 - Daily requirements of work (shifts) seven days a week
 - Weekdays hourly wage rate different from weekends
 - Worker may have different contracts
 - Workers' preferences of days-off
 - ...
- **Decision:** schedule workers to **meet** daily requirements, **satisfying** preferences, **minimising** weekly payroll.

ACK: Prof. Ender Özcan, UNUK, Prof. Tomas Maul & Dr Chen ZhiYuan, UNM



Example – The Solution I

- **Trial-and-error** with real system (experiments)
 - Learn from experimentation on the real system.
 - Not working if
 - Too many alternatives to explore.
 - Cost of making errors is very high.
 - Environment keeps changing.



Example – The Solution II

▪ Optimisation

- Model generates an optimal solution.
- Specify *precise mathematical relationships*.
- Limitations
 - Only work if the problem is structured and defined.
 - Quite often used to solve sub-problems.



Example – The Solution III

▪ Heuristics

- General *rules*
- *Informal knowledge* of how to solve problems efficiently and effectively, how to plan steps in solving a complex problem, how to improve performance, and so forth.



Example – The Solution IV

- **Simulation**

- Assume the appearance of the characteristics of reality.
 - Based on historical data using computer simulation tools.
 - “run the schedule”, compare costs, choose the best.

- **Issues**

- No guarantee that the solution is optimal one.
- Need professional development.



Decision Support using...

- **Trial-and-Error** is best for small-scale, hands-on problems where direct interaction is feasible and desirable.
- **Optimization** is ideal for well-defined problems with clear objectives and constraints, where finding the "best" solution is essential.
- **Heuristics** are practical for complex, poorly understood, or computationally intractable problems where quick, satisfactory solutions are sufficient.
- **Simulation** excels in situations where experimentation is too risky, expensive, or time-consuming.



Criterion	Trial-and-Error	Simulation	Optimization	Heuristics
Cost	High (equipment, resources, downtime)	Moderate (requires model setup but no physical resources)	Moderate to High (depends on algorithm complexity)	Low to Moderate (simple to implement)
Time Efficiency	Slow (real-world limitations)	Fast once the model is developed	Fast for small problems, slower for large/complex ones	Very fast in practice
Accuracy	High (direct system interaction)	Medium to High (depends on model accuracy)	High (for well-posed problems)	Low to Medium (approximate solutions)
Scalability	Low (logistical and cost limitations)	High (can test many scenarios)	High (scales with computational resources)	High (limited by heuristic quality)
Risk	High (potential damage or failure)	Low (virtual environment)	Low to Moderate (depends on constraints)	Low (minimal testing requirements)
Adaptability	Medium (reacts to real-world feedback)	High (can model dynamic systems)	Low to Medium (requires re-optimization)	High (rules can be adjusted easily)
Ease of Implementation	Low to Moderate (requires access to the actual system)	Medium (requires expertise to develop a realistic model)	Low to Moderate (needs mathematical modeling)	High (relies on domain knowledge and experience)
Optimality	Low to Medium (dependent on trials performed)	Medium (depends on model fidelity and scenarios tested)	High (designed to find the best solution)	Medium (focuses on "good enough" solutions)
Use Case Examples	Testing hardware configurations; tuning a physical system	Weather forecasting; testing car crash safety	Supply chain optimization; route planning	Scheduling; solving NP-hard problems (e.g., TSP)

ACK: Prof. Ender Özcan, UNUK, Dr Tomas Maul & Dr Chen Zhi



Summary



1. Types of Knowledge
2. Representation Methods
3. Case-Based Reasoning
4. Decision Making, Decision Support
5. **LLMs in Knowledge Representation and Reasoning**



ACK: Prof. Ender Özcan, UNUK, Prof. Tomas Maul & Dr Chen ZhiYuan, UNM



University of
Nottingham

UK | CHINA | MALAYSIA

NEXT:

**Modelling and
Simulation**